

NOTES ON THE LONG TIME DERIVATION OF BOLTZMANN'S EQUATION

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ABSTRACT. The purpose of these notes is to expose the structure, general strategy, and the main ideas of the proof in [3]. The core part of the proof is a purely combinatorial cutting algorithm for certain graphs called molecules. We first explain the steps reducing to this algorithm, and then explain the algorithm. These parts are essentially independent of each other.

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We give an exposition of the ideas and proof in [3]. The overall strategy consists of (a) proof of two key formulas (KF1)–(KF2); (b) reduction (using the two key formulas) to a purely combinatorial cutting algorithm for a certain class of graphs called molecules; and (c) construction of the cutting algorithm. These parts are essentially independent of each other, and we will explain each of them in these notes.

Before starting, we make some remarks about these notes:

- For the purpose of exposition and explaining the core ideas in an intuitive manner, and to avoid getting stuck in tedious notations and definitions, we will often use simplified definitions and consider simplified scenarios, compared to the actual proof. We will remark, when appropriate, on the extra complications that will occur in the real paper [3].
- In Section 2, we are presenting the proof of (KF1)–(KF2) in a superficially different way

from the real paper [3]. Namely, instead of introducing the interaction diagrams as an intermediate step, we introduce the molecules $\mathbb{M}$ directly (which are constructed as topological reductions of particle trajectories), and write indicator functions such as $\mathbb{1}_{\mathbb{M} \sim \mathbb{M}'}$ instead of $\mathbb{1}_{\lambda \sim \lambda'}$. In essence, this is still the same as the proof in [3], but this is simpler to present, and is closer to the formulation we will adopt in an upcoming revision of the paper.

- In Section 4, the discussion of the cutting algorithm is brief. The reader interested in this part should consult [4], which is the companion to these notes, and discusses a simplified 2-layer case of the algorithm, with full definitions and proofs. The notes in [4] will also be added to the upcoming revision as a new section “Toy models and the algorithm in simplified case”.
- We have included 18 figures to help explain the concepts and constructions. We opted to place those figures at the end of the notes in order not to disturb the continuity of the text.

1. STRUCTURE OF THE PROOF AND THE OUTLINE OF THE NOTES

In this section, we give a quick outline of the notes, which exposes the general structure of the proof. More precise explanations and definitions are often left to the relevant later sections.

Recall from [3], Section 1 that we are using the grand canonical ensemble with initial density $W_{0,N}$. The starting point is the following s -point marginal (or correlation function)

$$f_s(t, \mathbf{z}_s) := \varepsilon^{(d-1)s} \sum_{n=0}^{\infty} \frac{1}{n!} \int_{\mathbb{R}^{2dn}} W_{s+n}(t, \mathbf{z}_{s+n}) d\mathbf{z}_{s+1} \cdots d\mathbf{z}_{s+n}, \quad (1.1)$$

$$W_N(t, \mathbf{z}_N) := (\mathcal{S}_N(t)W_{0,N})(\mathbf{z}_N).$$

Here we abbreviate $\mathbf{z}_s = (z_1, \dots, z_s) \in \mathbb{R}^{2ds}$ (same for $\mathbf{z}_{s+n}$ and $\mathbf{z}_N$), and $\mathcal{S}_N(t)$ is the (transport) flow operator associated with the N -particle hard sphere system.

Recall also that we are splitting the long time interval as

$$[0, t_{\text{fin}}] = \bigcup_{\ell=1}^{\mathfrak{L}} [(\ell-1)\tau, \ell\tau],$$

where $0 < \tau \ll 1$ is appropriately chosen, so that the solution $n(t, z)$ of the Boltzmann equation on each interval $[(\ell-1)\tau, \ell\tau]$ admits a convergent power series expansion with base time $(\ell-1)\tau$. This choice of τ is made explicit in [3], Section 11, where we perform the analysis of $f^{\mathcal{A}}$ (see also Section 5); for now we only need it to be sufficiently small.

1.1. Key formulas for f_s and cumulants. In the first part of the proof, the goal is to prove the following two formulas for the marginals $f_s(t, \mathbf{z}_s)$ at times $t = \ell\tau$ ($0 \leq \ell \leq \mathfrak{L}$):

$$f_s(\ell\tau, \mathbf{z}_s) = \sum_{H \subseteq [s]} [f^{\mathcal{A}}(\ell\tau, \cdot)]^{\otimes ([s] \setminus H)} \cdot E_H(\ell\tau, \mathbf{z}_H) + \text{Err}(\ell\tau, \mathbf{z}_s), \quad (\text{KF1})$$

$$\|E_H(\ell\tau)\|_{L^1(\mathbb{R}^{2d|H|})} \lesssim \varepsilon^{(d-1)|H|} \sum_{\mathbb{M}} \mathcal{I}(\mathbb{M}), \quad (\text{KF2})$$

for $s \leq A_\ell := \lfloor \log \varepsilon \rfloor^{4^{\mathfrak{L}-\ell}}$ (note that $A_{\ell-1} = A_\ell^4$). We start by explaining these formulas below.

First, in (KF1) we denote $[s] := \{1, \dots, s\}$ and $\mathbf{z}_H = (z_j : j \in H)$, so (KF1) decomposes f_s into the factorized part $(f^{\mathcal{A}})^{\otimes}$ and the non-factorized part E_H up to error terms, at time $\ell\tau$. The factorized part $f^{\mathcal{A}}(\ell\tau) = f^{\mathcal{A}}(\ell\tau, z)$ is a one-point function which is an approximate solution to the Boltzmann equation, and the non-factorized part $E_H(\ell\tau) = E_H(\ell\tau, \mathbf{z}_H)$ is called the **cumulant**, which plays the central role in our proof.

Next, in (KF2) we decompose (an upper bound of) E_H into sums of integral terms over **molecules** $\mathbb{M}$; when measuring in the L^1_z norm, this allows to bound $\|E_H(\ell\tau)\|_{L^1}$ by sums of integral terms of form $\mathcal{I}(\mathbb{M})$ over molecules $\mathbb{M}$. The molecule $\mathbb{M}$, which records the collision history, is a fundamental concept in this paper, and will be discussed in depth in the sections below. The exact formula for $\mathcal{I}(\mathbb{M})$ will be given in (2.16) in Section 2.2; roughly speaking, $\mathcal{I}(\mathbb{M})$ is an integral expression that represents the (normalized) probability that the collision history encoded by $\mathbb{M}$ occurs.

The proof of (KF1)–(KF2) is by induction on ℓ . Suppose (KF1)–(KF2) is true for time $(\ell - 1)\tau$ and all $s \leq A_{\ell-1}$ (the case $\ell = 1$ being trivial), we shall prove it for time $\ell\tau$ and all $s \leq A_\ell$ (recall $A_{\ell-1} = A_\ell^4$). Roughly speaking, this includes the following steps; for details see Section 2.

- Specify all the collisions (and “virtual collisions” or overlaps) that are (directly or indirectly) related to the s particles in $f_s(\ell\tau)$, by performing cluster expansions and locating the collisions within each cluster and overlaps between clusters; in this way we produce a **molecule** $\mathbb{M}_\ell$ in **layer** ℓ consisting of C-atoms (for collisions) and O-atoms (for overlaps), see Definition 2.2.
- Consider all the particles involved in the collisions and overlaps described above; let their number be $s^+ \geq s$. We will perform a **truncation** to ensure $s^+ \leq A_\ell^4 = A_{\ell-1}$ (this is done by restricting the number of clusters, and the number of particles in each cluster, to be $\leq \Lambda_\ell := A_\ell^2$); for technical reasons, we also use truncations to ensure the number of recollisions in each cluster to be bounded by an absolute constant Γ .
- Now plug in the s^+ particle marginal at time $(\ell - 1)\tau$, which is expressed as a sum over molecules $\mathbb{M}'$ using the induction hypothesis (KF1)–(KF2). We then join $\mathbb{M}'$ with $\mathbb{M}_\ell$ to form a new molecule $\mathbb{M}$ (Section 2.1.6), and get a sum over $\mathbb{M}$. By isolating suitable connected components of $\mathbb{M}$, we can identify the factorized and cumulant parts of $f_s(\ell\tau)$; this leads to the expressions of $f^A(\ell\tau)$ and $E_H(\ell\tau)$, which will satisfy (KF1)–(KF2) for $\ell\tau$.

Finally, the error terms in (KF1) comes from two sources: first is produced by the truncation in layer ℓ described above, and second comes from a similar truncation in the cluster expansion. We shall discuss the first, more important, error term further in the concluding Section 5.

The role of molecules. In the above induction process, $\mathbb{M}'$ records the collision history (of suitable sets of particles) in $[0, (\ell - 1)\tau]$, and $\mathbb{M}_\ell$ records the collision history in $[(\ell - 1)\tau, \ell\tau]$, so $\mathbb{M} = \mathbb{M}' \cup \mathbb{M}_\ell$ records the collision history on $[0, \ell\tau]$. Intuitively, this molecule $\mathbb{M}$ is a **topological reduction** of the actual particle trajectories. More precisely (see Definition 2.2):

- The **collisions** and **overlaps** are represented by the **C-atoms** and **O-atoms** in $\mathbb{M}$, which are **abstract atoms**;
- The **linear pieces** of the trajectories (between two adjacent collisions or overlaps) are represented by the **edges** in $\mathbb{M}$, which are either abstract **bonds** connecting two atoms, or abstract **ends** sticking to one atom;
- The trajectory of each **particle** is represented by a set of atoms and bonds forming a **particle line**.

In the proof of (KF1)–(KF2) (Section 2), these intuitions will be very important in understanding the cluster expansions and how they translate to the molecule language. For example, the collisions within each **cluster** are just the C-atoms forming a **C-molecule** (Definition 2.1), and the overlaps between the clusters are just the O-atoms joining the C-molecules, see Definition 2.2.

However, we must emphasize that, the above correspondence will **not** be useful in the second and third parts of the proof. This is because these proofs are based on the key notion of **cutting**, which instantly breaks the above correspondence. We can still define cutting at the level of (abstract) molecules, where cutting a molecule reduces it into two smaller molecules, but they

will no longer correspond to the physical picture of particles, collisions and trajectories.

As such, in order to understand the proof in the second and third parts (Sections 3–4), it is important to think of molecules as **abstract combinatorial structures**, and **forget** all the notions of particles, collisions and trajectories etc. as well as the above correspondence, and maintain only those notions associated to molecules, such as bonds, atoms and particle lines etc.

1.2. Reduction to the algorithm. With the key formulas (KF1)–(KF2) established, it is then clear that our main theorem would follow from two facts:

- That f^A is close to the actual solution to Boltzmann’s equation;
- That $\|E_H\|_{L^1}$ is vanishing for any fixed $H \neq \emptyset$ (we understand that $E_\emptyset \equiv 1$).

The first fact is based on comparing f^A to the Duhamel expansion of Boltzmann’s equation, which will be briefly discussed in Section 5. The second fact occupies the main body of the proof. This is based on the following ingredients, which we discuss in Section 3:

- (1) *Bounding the number of terms in the sum* (KF2). Given any molecule $\mathbb{M}$, define the *circuit rank* ρ to be the number of (independent) cycles (this is zero if $\mathbb{M}$ is a tree, and increase by 1 for each new cycle created). We will prove that the number of molecules $\mathbb{M}$, with a given value of ρ , is bounded by

$$\boxed{\#\mathbb{M} \lesssim C^m |\log \varepsilon|^{C\rho}}, \quad (\text{KF3})$$

where $m = |\mathbb{M}|$ is the number of atoms in $\mathbb{M}$. We will justify this in Section 3.1 below.

- (2) *The cutting operation.* With the upper bound (KF3), the estimate of $\|E_H\|_{L^1}$ is then reduced to estimating $\mathcal{I}(\mathbb{M})$ for each individual $\mathbb{M}$. For this we will need to perform a **cutting** operation (see Section 3.2 for definition), to reduce $\mathbb{M}$ to smaller molecules. In terms of the integral $\mathcal{I}(\mathbb{M})$, we have the following: if $\mathbb{M}$ is cut into two molecules $\mathbb{M}_1$ and $\mathbb{M}_2$, then we essentially have

$$\boxed{\mathcal{I}(\mathbb{M}) \leq \mathcal{I}(\mathbb{M}_1)\mathcal{I}(\mathbb{M}_2)}. \quad (\text{KF4})$$

The more precise formula will be presented in Section 3 below, and it involves taking supremum over fixed ends for the $\mathcal{I}(\mathbb{M}_2)$ factor. This formula follows directly from Fubini’s theorem once we set up the notion of cutting (including fixed ends) properly. We will explain all this in Section 3.2 below.

Once (KF4) is established, we can iteratively apply it to cut $\mathbb{M}$ into a finite number of sub-molecules $\mathbb{M}_j$, each of which containing 1 or 2 atoms. The values of $\mathcal{I}(\mathbb{M}_j)$ can then be estimated explicitly, and from (KF4) we have that

$$\mathcal{I}(\mathbb{M}) \lesssim \prod_j \mathcal{I}(\mathbb{M}_j). \quad (1.2)$$

These molecules $\mathbb{M}_j$ are called *elementary molecules*. The precise definitions and estimates will be postponed to Section 3.3 below, but these elementary molecules can be split into three groups:

- (i) *Bad molecules:* The only such molecule is what we call a 4-molecule, which has one atom of degree 4 (i.e. one atom with four free ends). For such molecules $\mathbb{M}_j$, we have the lossy estimate $\mathcal{I}(\mathbb{M}_j) \lesssim \varepsilon^{-(d-1)}$, so one should try to minimize their number¹ in order to obtain sharper estimates on $\mathcal{I}(\mathbb{M})$.

¹Note the pre-factor $\varepsilon^{(d-1)|H|}$ in (KF2), which means we have the room to allow for at most $|H|$ bad molecules; this is useful when $|H|$ is relatively large. Since the main difficulty occurs in the case when $|H|$ is small, we will ignore this point from now on.

- (ii) *Normal molecules*: These are molecules such that² $\mathcal{I}(\mathbb{M}_j) \lesssim \tau$, i.e. without gain or loss; this includes the 3-molecule, which consists of one atom with three free ends, and one fixed end (we say it has degree 3, or **deg** 3).
- (iii) *Good molecules*: These are molecules such that $\mathcal{I}(\mathbb{M}_j) \lesssim \varepsilon$ (or a positive power of ε). The main example of those are $\{33\}$ molecules, which consist of two connected atoms, each having deg 3.

1.3. The algorithm. With all the above preparations, we can finally reduce the proof of our main theorem to the following **pure combinatorial estimate**:

- There exists a cutting sequence which cuts $\mathbb{M}$ into elementary molecules, such that

$$\boxed{\#\{33\} - 10d \cdot \#\{4\} \gtrsim \rho.} \quad (\text{KF5})$$

In fact, note that $\{4\}$ molecules are bad and $\{33\}$ molecules are good, and the others are normal. If we have (KF5), then using (KF2), (KF3) and (1.2), as well as the elementary molecule estimates in (i)–(iii) in Section 1.2 above, we get, denoting by $\mathfrak{M}_{m,\rho}$ the set of molecules with m atoms and circuit rank ρ , that

$$\begin{aligned} \|E_H\|_{L^1} &\lesssim \sum_{m,\rho \geq 1} \sum_{\mathbb{M} \in \mathfrak{M}_{m,\rho}} \mathcal{I}(\mathbb{M}), \\ \sum_{\mathbb{M} \in \mathfrak{M}_{m,\rho}} \mathcal{I}(\mathbb{M}) &\leq \#\mathbb{M} \cdot \sup_{\mathbb{M}} (\mathcal{I}(\mathbb{M})) \lesssim C^m |\log \varepsilon|^{C\rho} \cdot \prod_j \mathcal{I}(\mathbb{M}_j) \\ &\lesssim C^m |\log \varepsilon|^{C\rho} \cdot \tau^m \varepsilon^{-(d-1)\#\{4\} + \#\{33\}} \lesssim (C\tau)^m \varepsilon^{c\rho} |\log \varepsilon|^{C\rho}, \end{aligned}$$

which converges upon summing in $m, \rho \geq 1$, and is bounded by $\varepsilon^c \rightarrow 0$ for some constant $c > 0$. As explained above, this is all we need to prove the main theorem. In the paper [3], this argument appears in the *Proof of Proposition 5.3 assuming Proposition 6.15* which is located right after the statement of Proposition 6.15 in [3].

Therefore, it now suffices to construct a cutting sequence that satisfies the inequality (KF5). This is the goal of the third part of the proof (see [3], Sections 10–11). We discuss this construction, based on a number of *cutting algorithms*, in Section 4.

2. PART A: KEY FORMULAS (KF1) AND (KF2) AND THEIR PROOFS

We now discuss the proof of (KF1)–(KF2), with the quantity $\mathcal{I}(\mathbb{M})$ defined in (2.16) in Section 2.2. The proof consists of two parts: In the first part (Section 2.1), we prove (KF1) and an explicit molecule expansion of E_H (see (2.15)) by induction, and in the second part (Section 2.2), we prove that this formula for E_H implies (2.16). The notion of molecules $\mathbb{M}$, which will be introduced in the proof in Section 2.1, consists of multiple ingredients (such as C-atoms, O-atoms, layers etc.) We will gradually introduce them, starting from the basic C-molecules (Definition 2.1), then extending to the single-layer molecules with C- and O-atoms (Definition 2.2), and multi-layer molecules, based on the new ingredients we will need in the expansion process.

2.1. The cluster and molecule expansions. We start with justifying the key formula (KF1) together with the molecule expansion of E_H (see (2.15) below), which we do in steps. We remark that the arguments are heavily influenced by those in [5, 2, 1].

²This τ factor is due to the restriction that each collision belongs to a fixed time layer $[(\ell-1)\tau, \ell\tau]$, i.e. an interval of length τ . The same factors may also be put in (i) and (iii), but they are less important as τ is negligible compared to any power of ε . Below we will not worry about powers of τ as they never become a problem in the proof.

2.1.1. Prerequisite: truncated dynamics. Before starting the inductive proof, we first introduce the truncated dynamics which will be performed in each layer. The reason for this truncation is clear: it is difficult to estimate collision diagrams if there exist too large clusters or too many recollisions. As such, given $\Lambda = \Lambda_\ell := A_\ell^2$ (where $A_\ell = |\log \varepsilon|^{4^{2-\ell}}$, see Section 1.1) and a large constant Γ , we define the (Λ_ℓ, Γ) **truncated dynamics** on the time layer $[(\ell - 1)\tau, \ell\tau]$ as follows:

- The (Λ_ℓ, Γ) truncated dynamics is defined by **forcing** at any time that (a) the size of each cluster is $\leq \Lambda_\ell$, and (b) the number of recollisions in each cluster is $\leq \Gamma$.

In other words, if two particles form a “pre-collisional configuration” (i.e. reach distance ε after having distance $> \varepsilon$) and if allowing them to collide **would not violate** (a) or (b), then they will collide; otherwise, they are not allowed to collide and will simply pass through each other in straight lines.

Here the notion of **clusters** at any time t is defined (as usual) as the maximal set of particles that are connected to each other by collisions before t . Also, the number of **recollisions** in a cluster λ is defined as $\chi := m - |\lambda| + 1$ where m is the number of collisions within λ and before time t (this is the number of collisions beyond those necessary to make λ connected).

With the above definition of truncated dynamics, we can then decompose

$$f_s(\ell\tau) = \tilde{f}_s(\ell\tau) + f_s^{\text{err}}(\ell\tau), \quad (2.1)$$

where $\tilde{f}_s$ is the same as f_s in (1.1) but with the flow operator $\mathcal{S}_N(t)$ replaced by $\tilde{\mathcal{S}}_N(t)$ which is the flow operator³ associated with the new dynamics where we perform the (Λ_ℓ, Γ) truncation on $[(\ell - 1)\tau, \ell\tau]$ for each ℓ ; more precisely

$$\tilde{\mathcal{S}}_N(\ell\tau) = \mathcal{S}_N^\ell(\tau) \circ \mathcal{S}_N^{\ell-1}(\tau) \cdots \circ \mathcal{S}_N^1(\tau)$$

where $\mathcal{S}_N^\ell(\tau)$ is the (Λ_ℓ, Γ) truncated flow operator described above. The error term f_s^{err} produced by this truncation will go into the error term Err in (KF1) and requires a separate treatment, which will be discussed in Section 5. For this section we focus on the study of the main term $\tilde{f}_s$.

2.1.2. Step 1: Cluster expansions and C-molecules. We start by performing the cluster expansion⁴, which naturally leads to the notion of **C-molecules**. This is the simplest kind of molecules that only contains C-atoms (for collisions), which occur as topological reductions/abstractions of particle trajectories, counting only collisions. Intuitively, each collision is turned into an abstract C-atom, and each linear piece of trajectory (between two consecutive collisions, or before the first or after the last collision) is turned into an abstract edge (either a bond or bottom end or top end).

The formal definition of C-molecules is as follows, see Figure 1 for an illustration:

Definition 2.1 (C-molecules). A **C-molecule** is a graph whose vertices are called **C-atoms**. We also have **edges** which include **bonds** connecting two atoms (which are called **top edge** at the lower atom and **bottom edge** at the higher atom; note one of the two atoms is drawn higher than the other), and **ends** at one atom only (which may also be top or bottom). Each atom has exactly 2 top edges and 2 bottom edges (see the picture on the right in Figure 1). We divide the set of all edges into finitely many disjoint chains, such that each chain starts from a bottom end and keep going up until reaching a top end. Each such chain is called a **particle line**; for each atom $\mathbf{n}$ and each top edge and bottom edge at $\mathbf{n}$, we say they are **serial** if they belong to the same particle line.

³Note this truncated dynamics is no longer bijective, so the flow operator should be defined by a summation over all pre-images, which still preserves the L^1 norm, see [3], Definition 3.1.

⁴What we do here is superficially different from the real paper [3], in that we directly introduce the molecule picture without defining the interaction diagrams; however the arguments are in essence the same. We believe this is simpler and more intuitive; in fact we plan to adopt a similar approach, avoiding the interaction diagrams, in the revision of the paper.

With Definition 2.1, we can now construct the C-molecule associated with any given picture of particle trajectories (see Figure 1); this demonstrates how the C-molecules naturally occur as topological reductions, with those notions in the molecule picture naturally corresponding to those in the particle picture.

Given an initial configuration $\mathbf{z}^0 = \mathbf{z}_N^0$ and the associated particle trajectories under the (Λ_ℓ, Γ) truncated dynamics (see Section 2.1.1), we construct a C-molecule $\mathbb{M}$ as follows:

- Each (C-) atom in $\mathbb{M}$ is a collision between particles;
- Each bond (or bottom end, or top end) is a linear piece of trajectory of a particle between two consecutive collisions (or before the first collision, or after the last collision);
- For each bond between two collisions, the collision that happens later is drawn as the higher atom, and the bond is viewed as top/bottom at the lower/higher atom (top/bottom edges at an atom correspond to the post-collisional/pre-collisional configurations);
- Each particle line is the collection of all collisions and trajectory parts/edges for a specific particle (all the particle lines are in on-to-one correspondence with particles);
- Two edges at the same atom are serial if and only if they belong to the same particle trajectory.

The reader can verify the above correspondence in Figure 1. We make the following important remark here: At this point, and throughout this section, we will keep this one-to-one correspondence between particle trajectories and the molecules we construct. For example, we will sometimes refer to particles and particle lines as the same object. This identification will be stopped in Section 3 and on, when we start defining cutting operations on molecules, at which point this correspondence breaks and ceases to be useful.

The whole molecule $\mathbb{M}$ defined above is decomposed into connected components (see Figure 2), and each component (C-molecule) $\mathbb{M}_j$ consists of all collisions that happens in a given cluster λ_j (here a cluster again means maximal sets of particles connected by collisions). This gives a one-to-one correspondence between the clusters and the connected components $\mathbb{M}_j$ of $\mathbb{M}$. Again, we will abuse notation and often refer to $\mathbb{M}_j$ as the clusters for simplicity, and will say a particle belongs to $\mathbb{M}_j$ if it belongs to the corresponding λ_j (strictly speaking, one should say that the particle line corresponding to this particle is part of $\mathbb{M}_j$).

Now, given an initial configuration $\mathbf{z}_N^0$, we can find a unique molecule $\mathbb{M}$ with components $\mathbb{M}_j$ by the above topological reduction; conversely, given a molecule $\mathbb{M}$ with components $\mathbb{M}_j$ (as abstract graphs), the trajectories for $\mathbf{z}_N^0$ topologically reduces to $\mathbb{M}$ (meaning that the molecule constructed as above from the particle trajectories is exactly $\mathbb{M}$), if and only if

- All the particles in $\mathbb{M}_j$ are connected via collisions, and their trajectories topologically reduce to $\mathbb{M}_j$;
- For any $j \neq j'$, no collision happens between particles in $\mathbb{M}_j$ and $\mathbb{M}_{j'}$.

We define the indicator functions $\mathbb{1}_{\mathbb{M}_j}$ and $\mathbb{1}_{\mathbb{M}_j \not\sim \mathbb{M}_{j'}}$ for the above two statements respectively (which are functions of the initial configurations $\mathbf{z}_N^0$). Note that $\mathbb{1}_{\mathbb{M}_j}$ and $\mathbb{1}_{\mathbb{M}_j \not\sim \mathbb{M}_{j'}}$ actually only depend on the initial configurations of particles in $\mathbb{M}_j$ and $\mathbb{M}_j \cup \mathbb{M}_{j'}$ respectively. As a result, we have the main result of this subsection, namely the **cluster partition of unity**:

$$1 = \sum_{k \leq N} \sum_{\{\mathbb{M}_1, \dots, \mathbb{M}_k\}} \left(\prod_{1 \leq j < j' \leq k} \mathbb{1}_{\mathbb{M}_j \not\sim \mathbb{M}_{j'}} \right) \left(\prod_{j=1}^k \mathbb{1}_{\mathbb{M}_j} \right), \quad (2.2)$$

where the summation is taken over all $k \leq N$ and all disjoint molecules $(\mathbb{M}_j)_{j \leq k}$; with all the above preparations, (2.2) is now just a tautology.

We remark that, this version of cluster expansion (2.2), which is expressed in terms of molecules,

not only fixes the cluster of particles as sets, but also fixes the sets of collisions, and their order, within each cluster. We believe this is more convenient in terms of proving (KF2), as we will not need to introduce the intermediate concept of interaction diagrams.

Now, starting from (2.2), and noticing that $\widetilde{W}_N(\ell\tau) = \mathcal{S}_N^\ell(\tau)\widetilde{W}_N((\ell-1)\tau)$ where $\mathcal{S}_N^\ell(\tau)$ is the time τ flow operator associated with the (Λ_ℓ, Γ) truncated dynamics (due to the separate truncations in each layer in the definition of $\widetilde{W}_N$, see Section 2.1.1), we have

$$\widetilde{W}_N(\ell\tau, \mathbf{z}_N) = \sum_{k \leq N} \sum_{\{\mathbb{M}_1, \dots, \mathbb{M}_k\}} \left(\prod_{j=1}^k \mathcal{S}_{\mathbb{M}_j}(\tau) \cdot \mathbb{1}_{\mathbb{M}_j} \right) \left(\prod_{1 \leq j < j' \leq k} \mathbb{1}_{\mathbb{M}_j \not\sim \mathbb{M}_{j'}} \right) \widetilde{W}_N((\ell-1)\tau, \mathbf{z}_N), \quad (2.3)$$

where $\mathcal{S}_{\mathbb{M}_j}(\tau)$ is the time τ flow operator for the **partial dynamics** in $\mathbb{M}_j$ where we neglect all particles and collisions not in $\mathbb{M}_j$.

By plugging (2.3) into (1.1), we get a cluster expansion of $\widetilde{f}_s(\ell\tau)$ in terms of $\widetilde{W}_N((\ell-1)\tau)$, which currently still involves all $N = s + n$ particles. Eventually we will perform a resummation (Section 2.1.4) to reduce it back to $\widetilde{f}_{s^+}((\ell-1)\tau)$ that involve only the s^+ particles that are (directly or indirectly) related to the s particles in $\widetilde{f}_s(\ell\tau)$. To distinguish them, we define these s particles as **root particles** and label them by $[s] = \{1, \dots, s\}$; similarly, define a C-molecule $\mathbb{M}_j$ to be **root C-molecule** (corresponding to root clusters) if it contains a root particle, and let their number be $s_c \leq s$ (a root C-molecule may contain more than one root particle).

2.1.3. Step 2: O-atoms and general molecules. Starting from (2.3), we will perform the Penrose argument for the product $\prod_{1 \leq j < j' \leq k} \mathbb{1}_{\mathbb{M}_j \not\sim \mathbb{M}_{j'}}$ in Section 2.1.4 below. This would naturally lead to the notion of **overlaps**, **O-atoms**, and the general notion of molecules with C- and O-atoms; the purpose of this subsection is to explain these notions and the constructions leading to them. Roughly speaking, including the overlaps between different clusters, corresponds to **joining** two C-molecules by an O-atom along two suitable edges (Definition 2.2 and Figure 3).

Define $\mathbb{1}_{\mathbb{M}_j \sim \mathbb{M}_{j'}} := 1 - \mathbb{1}_{\mathbb{M}_j \not\sim \mathbb{M}_{j'}}$, which is the opposite to the indicator function defined in Section 2.1.2. Let us start with a simple example with only two clusters $\mathbb{M}_1$ and $\mathbb{M}_2$ in (2.3); we have

$$\begin{aligned} & (\mathcal{S}_{\mathbb{M}_1}(\tau) \cdot \mathbb{1}_{\mathbb{M}_1})(\mathcal{S}_{\mathbb{M}_2}(\tau) \cdot \mathbb{1}_{\mathbb{M}_2}) \cdot \mathbb{1}_{\mathbb{M}_1 \not\sim \mathbb{M}_2} \\ &= (\mathcal{S}_{\mathbb{M}_1}(\tau) \cdot \mathbb{1}_{\mathbb{M}_1})(\mathcal{S}_{\mathbb{M}_2}(\tau) \cdot \mathbb{1}_{\mathbb{M}_2}) - (\mathcal{S}_{\mathbb{M}_1}(\tau) \cdot \mathbb{1}_{\mathbb{M}_1})(\mathcal{S}_{\mathbb{M}_2}(\tau) \cdot \mathbb{1}_{\mathbb{M}_2}) \mathbb{1}_{\mathbb{M}_1 \sim \mathbb{M}_2}. \end{aligned}$$

The last term above is supported on the following set of initial configurations: In the “factorized” dynamics $\mathcal{S}_{\mathbb{M}_1}(\tau)\mathcal{S}_{\mathbb{M}_2}(\tau)$, there exists a time where a particle in $\mathbb{M}_1$ and a particle in $\mathbb{M}_2$ form a pre-collisional configuration, i.e. the trajectories of two particles from $\mathbb{M}_1$ and $\mathbb{M}_2$ **overlap**, and they pass through each other without changing velocity. Crucially, this overlap comes with a negative sign coming from the inclusion-exclusion relation above, which will eventually translate into the negative sign of the loss terms in Boltzmann’s equation.

As a result, the relation $\mathbb{1}_{\mathbb{M}_1 \sim \mathbb{M}_2}$ can then be written as a sum over possible overlaps of the trajectories of particles in $\mathbb{M}_1$ and $\mathbb{M}_2$. Just like collisions, these overlaps can be encoded in the language of molecules by introducing a new type of atoms, which we call O-atoms. This O-atom is the intersection of a two particle lines, one from $\mathbb{M}_1$ and another from $\mathbb{M}_2$ (see Figure 3). This leads us to introduce the general notion of molecules as follows:

Definition 2.2 (Molecules with O-atoms). We make the following definitions:

- A **molecule** $\mathbb{M}$ is just like a C-molecule in Definition 2.1, but with two types of atoms: C-atoms and O-atoms. All the other notations (edges, bonds, ends, particle lines etc.) are the same as Definition 2.1, see Figure 3.

- Starting from a C-molecule $\mathbb{M}$, given any two edges e_1 and e_2 in two distinct clusters $\mathbb{M}_1$ and $\mathbb{M}_2$ (i.e. connected components of $\mathbb{M}$), we can **join** e_1 and e_2 by adding an O-atom $\mathbf{n}$ as a point of intersection of the edges e_1 and e_2 as in [Figure 3](#), and draw it in the interior of e_1 and e_2 (for example, if e_j connects two atoms, then $\mathbf{n}$ should be drawn between these two atoms), to form a new molecule in which $\mathbb{M}_1$ and $\mathbb{M}_2$ are connected via this new O-atom. Note this construction can be repeated to connect other clusters.

Therefore, in connection with the particle trajectory picture, the C-atoms in $\mathbb{M}$ correspond to collisions within each $\mathbb{M}_j$, and O-atoms correspond to overlaps between particles in $\mathbb{M}_j$ and $\mathbb{M}_{j'}$ which join these clusters; particle lines are still in one-to-one correspondence with particles, see [Figures 3–4](#). In particular, this leads to the further decomposition of $\mathbb{1}_{\mathbb{M}_1 \sim \mathbb{M}_2}$ (corresponding to specifying which two particles overlap, and where this overlap is placed among the collisions):

$$\mathbb{1}_{\mathbb{M}_1 \sim \mathbb{M}_2} = \sum_{\mathbb{M}} \mathbb{1}_{\mathbb{M}}, \quad (2.4)$$

where the summation is taken over all molecules $\mathbb{M}$ formed by joining $\mathbb{M}_1$ to $\mathbb{M}_2$ with *one* O-atom.

From clusters to molecules and back: We have seen how, given a collection of C-molecules representing the clusters, we can form a general molecule $\mathbb{M}$ by joining those clusters with O-atoms. Conversely, given a molecule $\mathbb{M}$, the C-molecules and clusters can be recovered by deleting all the O-atoms (combining the two serial edges at each deleted O-atom into a new edge) and looking at the connected components of the resulting molecule. This allows us to introduce the following definition:

Definition 2.3 (Cluster graph of a molecule). Given a molecule $\mathbb{M}$ and consider its clusters $\mathbb{M}_j$ ($1 \leq j \leq k$) of C-molecules defined above. We will form a graph $G_{\mathbb{M}}$, called the **cluster graph of $\mathbb{M}$** , as follows: the vertices of $G_{\mathbb{M}}$ are the clusters $\{\mathbb{M}_1, \dots, \mathbb{M}_k\}$, and we connect two vertices $\mathbb{M}_j$ and $\mathbb{M}_{j'}$ in $G_{\mathbb{M}}$ if and only if they are joined by an O-atom in $\mathbb{M}$, see [Figure 4](#).

2.1.4. Step 3: The Penrose argument. Now, we will perform the full inclusion-exclusion expansion of the product $\prod_{1 \leq j < j' \leq k} \mathbb{1}_{\mathbb{M}_j \sim \mathbb{M}_{j'}}$ and apply the Penrose argument similar to that in [2, 1]. Recall the notion of root particles and root clusters (i.e. clusters containing at least one root particles) in Section 2.1.2; we denote the set of root particles (and corresponding root particle lines) by $r(\mathbb{M})$.

Now, starting from (2.3), the Penrose argument consists of first applying the inclusion-exclusion expansion to the product $\prod_{1 \leq j < j' \leq k} \mathbb{1}_{\mathbb{M}_j \sim \mathbb{M}_{j'}}$:

$$\prod_{1 \leq j < j' \leq k} \mathbb{1}_{\mathbb{M}_j \sim \mathbb{M}_{j'}} = \left(\prod_{1 \leq j < j' \leq k} (1 - \mathbb{1}_{\mathbb{M}_j \sim \mathbb{M}_{j'}}) \right) = \sum_G (-1)^{|E(G)|} \prod_{\{\mathbb{M}_j, \mathbb{M}_{j'}\} \in E(G)} \mathbb{1}_{\mathbb{M}_j \sim \mathbb{M}_{j'}} \quad (2.5)$$

(where the sum is taken over all simple graphs G with vertex set $\{\mathbb{M}_1, \dots, \mathbb{M}_k\}$, and $E(G)$ is the edge set of G), and then regrouping the terms in the summation (2.5) to reduce it to a smaller sum

$$(2.5) = \sum_F \tilde{\chi}_F, \quad (2.6)$$

where the sum in (2.6) is taken over all graphs F that are **forests** (i.e. without cycles) with vertex set being a **subset** of $\{\mathbb{M}_1, \dots, \mathbb{M}_k\}$. This is basically our **cluster forest** expansion, where the forest F will serve as the cluster graph $G_{\mathbb{M}}$ of the molecule $\mathbb{M}$ in layer ℓ , which is formed by joining the C-molecules (clusters) by the O-atoms (overlaps), see [Figure 4](#). For the proof of (2.6), we refer to either [3], Lemma 4.27 or [1, 2], where the arguments are essentially the same. For now, we only recall the following properties of F and $\tilde{\chi}_F$:

- F is a forest, whose vertex set is a subset of all clusters $\{\mathbb{M}_1, \dots, \mathbb{M}_k\}$ that contains all root clusters. Each connected component of F contains a root cluster;
- With a truncation, we can assume the number of clusters in F is bounded by Λ_ℓ (as well as the number of particles in each cluster due to the truncated dynamics); the case when this is violated will go into the error term Err in (KF1) and we will not discuss them (they are exceedingly small due to our choice of Λ_ℓ).
- We have

$$\tilde{\chi}_F = \chi_F \prod_{\mathbb{M}_j, \mathbb{M}_{j'} \notin F} \mathbb{1}_{\mathbb{M}_j \not\sim \mathbb{M}_{j'}}, \quad |\chi_F| \leq \prod_{\{\mathbb{M}_j, \mathbb{M}_{j'}\} \in E(F)} \mathbb{1}_{\mathbb{M}_j \sim \mathbb{M}_{j'}}, \quad (2.7)$$

where χ_F depends only on the initial configurations of the particles in clusters in F .

Resummation: The important thing in (2.7) is that we retain the $\prod_{j, j' \notin F} \mathbb{1}_{\mathbb{M}_j \not\sim \mathbb{M}_{j'}}$ factor. By plugging (2.5)–(2.6) into (2.3), we get

$$\widetilde{W}_N(\ell\tau, \mathbf{z}_N) = \sum_{k \leq N} \sum_{\{\mathbb{M}_1, \dots, \mathbb{M}_k\}} \sum_F \left(\prod_{j=1}^k \mathcal{S}_{\mathbb{M}_j}(\tau) \cdot \mathbb{1}_{\mathbb{M}_j} \right) \left(\chi_F \cdot \prod_{\mathbb{M}_j, \mathbb{M}_{j'} \notin F} \mathbb{1}_{\mathbb{M}_j \not\sim \mathbb{M}_{j'}} \right) \widetilde{W}_N((\ell-1)\tau, \mathbf{z}_N). \quad (2.8)$$

Note that in (2.8) we can extract the factor

$$\left(\prod_{\mathbb{M}_j \notin F} \mathcal{S}_{\mathbb{M}_j}(\tau) \cdot \mathbb{1}_{\mathbb{M}_j} \right) \left(\prod_{\mathbb{M}_j, \mathbb{M}_{j'} \notin F} \mathbb{1}_{\mathbb{M}_j \not\sim \mathbb{M}_{j'}} \right) \quad (2.9)$$

let the $\mathbb{M}_j$ that are not in F be $\{\mathbb{M}'_1, \dots, \mathbb{M}'_r\}$, then by applying the partition of unity (2.2) to the summation in $(\mathbb{M}'_1, \dots, \mathbb{M}'_r)$, we can rewrite the operator

$$\sum_r \sum_{\{\mathbb{M}'_1, \dots, \mathbb{M}'_r\}} \left(\prod_{j=1}^r \mathcal{S}_{\mathbb{M}'_j}(\tau) \cdot \mathbb{1}_{\mathbb{M}'_j} \right) \left(\prod_{1 \leq j < j' \leq r} \mathbb{1}_{\mathbb{M}'_j \not\sim \mathbb{M}'_{j'}} \right) \quad (2.10)$$

as the flow operator associated to the truncated dynamics for all the other particles not in one of the clusters in F . Since this flow operator preserves the L^1 norm, upon integrating, we can completely get rid of those particles, which replaces $\widetilde{W}_N((\ell-1)\tau)$ in (2.8) by $\widetilde{f}_{s^+}((\ell-1)\tau)$ where s^+ is the number of the particles in a cluster in F . As for those $\mathbb{M}_j \in F$, we use the upper bound⁵ of χ_F in (2.7), and then expand each $\mathbb{1}_{\mathbb{M}_j \sim \mathbb{M}_{j'}}$ factor using (2.4). By plugging all these into (1.1) and integrating in all the non-root-particle variables, we get

$$\begin{aligned} \widetilde{f}_s(\ell\tau, \mathbf{z}_s) &= \sum_{\mathbb{M}} \varepsilon^{-(d-1)(|p(\mathbb{M})|-s)} \int_{\mathbb{R}^{2d(|p(\mathbb{M})|-s)}} (\mathcal{S}_{\mathbb{M}}(\tau) \cdot \mathbb{1}_{\mathbb{M}}) f_{|p(\mathbb{M})|}((\ell-1)\tau, \mathbf{z}_{p(\mathbb{M})}) d\mathbf{z}_{p(\mathbb{M}) \setminus r(\mathbb{M})}, \\ \mathcal{S}_{\mathbb{M}}(\tau) &= \prod_j \mathcal{S}_{\mathbb{M}_j}(\tau). \end{aligned} \quad (2.11)$$

Here in (2.11) we have:

- $\mathbb{M}$ is the molecule formed by joining the C-molecules $\mathbb{M}_j \in F$ by O-atoms, such that the cluster graph $G_{\mathbb{M}} = F$ is exactly as in (2.6), see Figure 4. In other words, with fixed F , the clusters (equivalently C-molecules) $\mathbb{M}_j$ get joined by O-atoms in $\mathbb{M}$ exactly as prescribed by the graph F (i.e. we join $\mathbb{M}_j$ and $\mathbb{M}_{j'}$ by an O-atom if and only if $\mathbb{M}_j$ is connected to $\mathbb{M}_{j'}$ by an edge in $E(F)$). The sum over F in (2.8), and over the clusters $\mathbb{M}_j \in F$, is now transformed into a free summation over all $\mathbb{M}$ in (2.11). We will denote this $\mathbb{M}$ in (2.11) by

⁵Strictly speaking we should include χ_F in the formula; here, for the purpose of demonstration, we are pretending χ_F is equal to its upper bound in (2.7).

$\mathbb{M}_\ell$ in Section 2.1.6 below, as it represents the single layer molecule in layer ℓ .

- $p(\mathbb{M})$ is the set of all particles in $\mathbb{M}$ (i.e. all particles in clusters in F); we have denoted $|p(\mathbb{M})| := s^+$ above. On the other hand, $r(\mathbb{M})$ is the set of root particles, i.e. particles occurring in $f_s(\ell\tau, \mathbf{z}_s)$.
- $\mathcal{S}_{\mathbb{M}}(\tau)$ is defined as in (2.11), with the product taken over all C-molecules $\mathbb{M}_j$ forming $\mathbb{M}$ (which can be read from $\mathbb{M}$ as explained just before Definition 2.3). Finally, $\mathbb{1}_{\mathbb{M}}$ is the indicator function restricting to initial configurations for which the collisions and overlaps in the trajectory happen exactly as that described by $\mathbb{M}$.

2.1.5. *Step 4: Factorization and the single-layer expansion.* We start from (2.11). This formula allows us to express $f_s(\ell\tau, \mathbf{z}_s)$ by integrals of $f_{s^+}((\ell-1)\tau, \mathbf{z}_{p(\mathbb{M})})$, where $s^+ = |p(\mathbb{M})|$ and $p(\mathbb{M})$ is the set of all particles in $\mathbb{M}$ as above. By applying the induction hypothesis (KF1) for $(\ell-1)\tau$, namely

$$f_{|p(\mathbb{M})|}((\ell-1)\tau, \mathbf{z}_{p(\mathbb{M})}) = \sum_{H' \subseteq p(\mathbb{M})} (f^{\mathcal{A}}((\ell-1)\tau))^{\otimes(p(\mathbb{M}) \setminus H')} E_{H'}((\ell-1)\tau, \mathbf{z}_{H'}),$$

we get that

$$\begin{aligned} \tilde{f}_s(\ell\tau, \mathbf{z}_s) &= \sum_{\mathbb{M}} \sum_{H' \subseteq p(\mathbb{M})} \varepsilon^{(d-1)(s-|p(\mathbb{M})|)} \int_{\mathbb{R}^{2d(|p(\mathbb{M})|-s)}} (\mathcal{S}_{\mathbb{M}}(\tau) \cdot \mathbb{1}_{\mathbb{M}}) \\ &\quad \left[(f^{\mathcal{A}}((\ell-1)\tau))^{\otimes(p(\mathbb{M}) \setminus H')} E_{H'}((\ell-1)\tau) \right] d\mathbf{z}_{p(\mathbb{M}) \setminus r(\mathbb{M})}. \end{aligned} \quad (2.12)$$

The next question is: on the right hand side of (2.12), which is a function of $\mathbf{z}_s$, what single variable factors can we extract? Such single variable factors must satisfy three conditions: (i) it must correspond to a connected component of $\mathbb{M}$ (otherwise this variable is intertwined with others via $\mathbb{M}$), (ii) this component must contain only one root particle (otherwise we again have two intertwined root particles), and (iii) this component must not contain any particle occurring in H' (otherwise this is intertwined with others via $E_{H'}$). As such, we need to isolate all those components of $\mathbb{M}$ that satisfy (i)–(iii).

Let $(\mathbb{M}_\alpha)$ be all components of $\mathbb{M}$, then summing over $\mathbb{M}$ is equivalent to summing over all $\mathbb{M}_\alpha$. If some $\mathbb{M}_\alpha$ satisfies (i)–(iii) above, then this should contribute to the factorized $f^{\mathcal{A}}(\ell\tau)$ factor; moreover $r(\mathbb{M}_\alpha)$ will have a single element as $\mathbb{M}_\alpha$ contains only one root particle. Since the input only contains $f^{\mathcal{A}}((\ell-1)\tau)$ factors, and not $E_{H'}$ by (iii), we then get

$$f^{\mathcal{A}}(\ell\tau, \mathbf{z}) = \sum_{\mathbb{M}: |r(\mathbb{M})|=1} \varepsilon^{-(d-1)(|p(\mathbb{M})|-1)} \int_{\mathbb{R}^{2d(|p(\mathbb{M})|-1)}} (\mathcal{S}_{\mathbb{M}}(\tau) \mathbb{1}_{\mathbb{M}}) [f^{\mathcal{A}}((\ell-1)\tau)]^{\otimes p(\mathbb{M})} d\mathbf{z}_{p(\mathbb{M}) \setminus r(\mathbb{M})}. \quad (2.13)$$

This sum is further discussed in Section 5. As for the non-factorized cumulant part E_H , it involves those components of $\mathbb{M}$ that do not satisfy (i)–(iii), namely

$$\begin{aligned} E_H(\ell\tau, \mathbf{z}_H) &= \sum_{\mathbb{M}: r(\mathbb{M})=H}^{(*)} \varepsilon^{-(d-1)(|p(\mathbb{M})|-|H|)} \int_{\mathbb{R}^{2d(|p(\mathbb{M})|-|H|)}} (\mathcal{S}_{\mathbb{M}}(\tau) \mathbb{1}_{\mathbb{M}}) \\ &\quad \left((f^{\mathcal{A}}((\ell-1)\tau))^{\otimes(p(\mathbb{M}) \setminus H')} E_{H'}((\ell-1)\tau) \right) d\mathbf{z}_{p(\mathbb{M}) \setminus r(\mathbb{M})}. \end{aligned} \quad (2.14)$$

Here in (2.14), the summation $(*)$ is taken over all molecules $\mathbb{M}$ as above, such that each connected component of $\mathbb{M}$ either contains more than one root particle (i.e. those in $r(\mathbb{M}) = H$), or contains

a particle occurring in H' .

Remarks on the molecule $\mathbb{M}_\ell$ occurring in $E_H(\ell\tau)$: The single layer molecule $\mathbb{M}$ will be denoted by $\mathbb{M}_\ell$ below, as it will be connected to the molecule $\mathbb{M}'$ in layer $\ell - 1$ and lower (see next Section 2.1.6). We make a few remarks regarding this molecule $\mathbb{M}_\ell$:

- The root particle/root particle line set of $\mathbb{M}_\ell$ is $r(\mathbb{M}_\ell) = H$, which is a subset of the set $p(\mathbb{M}_\ell)$ of all particles in $\mathbb{M}_\ell$.
- There is another subset H' of $p(\mathbb{M})$ of particles corresponding to the cumulant $E_{H'}((\ell - 1)\tau)$ in (2.14). The particles in $p(\mathbb{M}) \setminus H'$ only occur in (2.14) via the f^A function; intuitively speaking, we will not care about their history before time $(\ell - 1)\tau$. On the other hand, those particles in H' will enter or “cross into” layers $(\ell - 1)$ and lower (i.e. time $(\ell - 1)\tau$ and before), via the molecule $\mathbb{M}'$ which occurs in the further expansion of $E_{H'}((\ell - 1)\tau)$ using the induction hypothesis (KF1). In fact, this set H' is exactly the set of root particles $r(\mathbb{M}')$ for the molecule $\mathbb{M}'$ of layer $\ell - 1$ and lower.
- The following statements hold within the molecule $\mathbb{M}_\ell$: (i) each connected component of $\mathbb{M}_\ell$ must intersect a particle line in $r(\mathbb{M}_\ell) = H$ (i.e. contains a particle in H), and (ii) each connected component of $\mathbb{M}_\ell$ either intersects more than one particle line in $r(\mathbb{M}_\ell)$, or intersects a particle line in $r(\mathbb{M}_{\ell-1}) := H'$ (i.e. contains a particle in H'). Here (i) is because each cluster is connected to a root cluster by a sequence of overlaps (cf. the property of the cluster graph F in Section 2.1.4), and (ii) is the requirement $(*)$ in the summation (2.14).

2.1.6. Step 5: Multi-layer expansion and multi-layer molecules. Now, with (2.14), which expresses $E_H(\ell\tau)$ in terms of f^A factors and $E_H((\ell - 1)\tau)$, we are finally ready to prove an inductive formula for E_H , which eventually leads to (KF2). We first present this formula below⁶:

$$E_H(\ell\tau, z_H) = \sum_{\mathbb{M}} \varepsilon^{-(d-1)(|p(\mathbb{M})| - |H|)} \int_{\mathbb{R}^{2d(|p(\mathbb{M})| - |H|)}} (\mathcal{S}_{\mathbb{M}}(\ell\tau) \cdot \mathbb{1}_{\mathbb{M}}) \left(\prod_{\mathbf{p} \in p(\mathbb{M})} f^A((\ell[\mathbf{p}] - 1)\tau, z_{\mathbf{p}}) \right) dz_{p(\mathbb{M}) \setminus r(\mathbb{M})}. \quad (2.15)$$

Here in (2.15):

- The sum is taken over all **multi-layer molecules** $\mathbb{M}$ which are formed by joining the molecules $\mathbb{M}_{\ell'}$ in layers $\ell' \leq \ell$, (which are just the analogs of $\mathbb{M} = \mathbb{M}_\ell$ in Section 2.1.5 for lower time layers, and they contain both C- and O-atoms); for precise definition, see discussion below.
- The $p(\mathbb{M})$ is the set of all particles involved in $\mathbb{M}$, and $r(\mathbb{M}) = H$ is still the set of root particles; the definition of the mapping $\ell[\mathbf{p}] \in [1, \ell]$ and the operator $\mathcal{S}_{\mathbb{M}}$ is defined in the discussion below.

Joining molecules from consecutive layers: The molecule $\mathbb{M}$ in (2.15) is formed by joining the $\mathbb{M}_\ell$ in Section 2.1.5 with molecules in layers $\ell - 1$ and lower (See Figure 5 for the illustration). This can be done in an inductive process: suppose we have $\mathbb{M}'$ which is formed by joining $(\mathbb{M}_{\ell-1}, \dots, \mathbb{M}_1)$, then we can join $\mathbb{M}_\ell$ with $\mathbb{M}'$ to form $\mathbb{M}$ as follows:

- As a pre-requisite for joining, we require that the set of root particles $r(\mathbb{M}') = H'$ where H' is as in Section 2.1.5 (as $\mathbb{M}'$ occurs in expanding $E_{H'}((\ell - 1)\tau)$, just as $\mathbb{M}_\ell$ occurs in expanding $E_H(\ell\tau)$).

⁶We are ignoring the initial cumulants at time $t = 0$ caused by the restrictions $|x_j - x_k| \geq \varepsilon$; in practice they are relatively trivial to deal with (compared with the main parts of the proof).

- For each particle in H' , it has a corresponding particle line in $\mathbb{M}_\ell$ (which contains a unique *bottom* free end), and a corresponding particle line in $\mathbb{M}'$ (which contains a unique *top* free end). We then merge these two ends to form a bond between the two atoms previously at these two ends. Physically, this corresponds to combining the trajectories of the same particle in H' in $[0, (\ell - 1)\tau]$ and $[(\ell - 1)\tau, \ell\tau]$ to get a full trajectory on $[0, \ell\tau]$ (cf. [Figure 5](#)).
- The $\mathbb{M}_\ell$ is called the ℓ -th **layer** of $\mathbb{M}$, and those $\mathbb{M}_{\ell'}$ ($\ell' \leq \ell - 1$) forming $\mathbb{M}'$ is called the ℓ' -th layer.
- The set of root particles of $\mathbb{M}$ is equal to that of the top layer $\mathbb{M}_\ell$, i.e. $r(\mathbb{M}) = r(\mathbb{M}_\ell) = H$; the set of root particles of $\mathbb{M}'$ is equal to that of the *next* layer $\mathbb{M}_{\ell-1}$ (the top layer of $\mathbb{M}'$), i.e. $r(\mathbb{M}') = r(\mathbb{M}_{\ell-1}) = H'$. In the same way we have the sets $r(\mathbb{M}_{\ell'})$ (and $p(\mathbb{M}_{\ell'})$ which is the set of all particles involved in $\mathbb{M}_{\ell'}$) for all $\ell' \leq \ell$.
- The mapping $\ell[\mathbf{p}]$ is defined for each particle $\mathbf{p}$ as follows: if $\mathbf{p} \in p(\mathbb{M}_\ell) \setminus r(\mathbb{M}_{\ell-1})$ (i.e. if $\mathbf{p}$ occurs in the layer ℓ expansion via one of the f^A factors, so does not enter into $\mathbb{M}'$) then define $\ell[\mathbf{p}] = \ell$; otherwise $\mathbf{p}$ occurs in some earlier layers in $\mathbb{M}'$, then define $\ell[\mathbf{p}]$ inductively in $\mathbb{M}'$. Heuristically, this number is the **earliest layer** that this particle can be traced back to (i.e. the earliest layer that it has collisions or overlaps in $\mathbb{M}$); for example $\ell[\mathbf{p}] = \ell$ if and only if $\mathbf{p}$ occurs in layer ℓ but does not occur in any earlier layers (hence not in $\mathbb{M}'$).
- Finally $\mathcal{S}_{\mathbb{M}}(\ell\tau)$ is (essentially) the flow operator associated with the flow prescribed by $\mathbb{M}$ (i.e. we ignore all particles and collisions not in $\mathbb{M}$), and $\mathbb{1}_{\mathbb{M}}$ is the indicator function that all collisions and overlaps in $\mathbb{M}$ do occur, as before. If $\mathbb{M}$ is formed by joining different layers $\mathbb{M}_{\ell'}$ ($1 \leq \ell' \leq \ell$), then up to trivial linear transports, we have $\mathcal{S}_{\mathbb{M}}(\ell\tau) = \mathcal{S}_{\mathbb{M}_\ell}(\tau) \circ \cdots \circ \mathcal{S}_{\mathbb{M}_1}(\tau)$ with $\mathcal{S}_{\mathbb{M}_{\ell'}}(\tau)$ defined in Section 2.1.4.

Proof of (2.15): Assuming (2.15) is true for $E_{H'}((\ell - 1)\tau)$, we plug it into (2.14). By combining the ε powers, the f^A factors, the $\mathcal{S}_{\mathbb{M}}$ flow operators and integrations associated with $\mathbb{M}'$ and with $\mathbb{M}_\ell$, we get the corresponding quantities for $\mathbb{M}$, which proves (2.15) for $E_H(\ell\tau)$. We only check here the statement for the ε power: note that $|p(\mathbb{M})| - |p(\mathbb{M}')| = |p(\mathbb{M}_\ell) \setminus H'|$ (i.e. a particle belongs to $\mathbb{M}$ and not to $\mathbb{M}'$, if and only if it occurs in $\mathbb{M}_\ell$ and does not cross into earlier layers via H'), so

$$(|p(\mathbb{M}')| - |H'|) + (|p(\mathbb{M}_\ell)| - |H|) = |p(\mathbb{M})| - |H|.$$

As a conclusion of all this, we have now proved (KF1) with f^A given by (2.13) and E_H given by (2.15). This completes the task of this subsection. In Section 2.2 we will use this to deduce (KF2).

Properties of $\mathbb{M}$: Before moving on, let us record some key properties of the multi-layered molecule $\mathbb{M}$ occurring in this subsection, which will be useful in the cutting algorithm. These properties directly follow from construction, or from the properties of the single layer molecules $\mathbb{M}_\ell$ in Section 2.1.5; see Figure 6.

- $\mathbb{M}$ contains ℓ layers $\mathbb{M}_{\ell'}$ ($1 \leq \ell' \leq \ell$). For each ℓ' there is a set $p(\mathbb{M}_{\ell'})$ of all particles/particle lines in $\mathbb{M}_{\ell'}$, and the set $r(\mathbb{M}_{\ell'})$ of all root particles/particle lines in $\mathbb{M}_{\ell'}$ (i.e. those crossing into layer ℓ' from higher layers). Define $s_{\ell'} = |r(\mathbb{M}_{\ell'})|$. As convention we write $r(\mathbb{M}_0) = \emptyset$.
- We have $r(\mathbb{M}_{\ell-1}) = p(\mathbb{M}_{\ell'}) \cap p(\mathbb{M}_{\ell-1})$.
- Every particle line in $p(\mathbb{M}_{\ell'})$ is connected to a particle line in $r(\mathbb{M}_{\ell'})$ within $\mathbb{M}_{\ell'}$.
- Every particle line in $r(\mathbb{M}_{\ell'})$ is either connected to another particle line in $r(\mathbb{M}_{\ell'})$, or to a particle line in $r(\mathbb{M}_{\ell-1})$, within $\mathbb{M}_{\ell'}$.

2.2. Key Formula (KF2) and the integral expression for $\mathcal{I}(\mathbb{M})$. In this subsection, starting from (2.15), we will justify the second key formula (KF2) with the following expression for $\mathcal{I}(\mathbb{M})$:

$$\begin{aligned} \mathcal{I}(\mathbb{M}) := & \varepsilon^{-(d-1)(|\mathcal{E}| - 2|\mathcal{M}|)} \int_{\mathbb{R}^{2d|\mathcal{E}|} \times \mathbb{R}^{|\mathcal{M}|}} \mathbb{1}_{\mathcal{D}}(\mathbf{t}_{\mathcal{M}}) \cdot \prod_{\mathbf{n} \in \mathcal{M}} \Delta(z_{e_1}, z_{e_2}, z_{e'_1}, z_{e'_2}, t_{\mathbf{n}}) \\ & \times \prod_{e \in \mathcal{E}_{\text{end}}^-} |f^{\mathcal{A}}((\ell[e] - 1)\tau, x_e - (\ell[e] - 1)\tau \cdot v_e, v_e)| \, d\mathbf{z}_{\mathcal{E}} d\mathbf{t}_{\mathcal{M}}. \end{aligned} \quad (2.16)$$

Here, $\mathcal{E}$ is the set of all edges of $\mathbb{M}$ and $\mathcal{M}$ is the set of atoms (sometimes we may abuse notations and write $\mathbb{M}$ instead of $\mathcal{M}$ and $|\mathbb{M}|$ instead of $|\mathcal{M}|$), and $\mathbf{z}_{\mathcal{E}} = (z_e : e \in \mathcal{E})$ and $\mathbf{t}_{\mathcal{M}} = (t_{\mathbf{n}} : \mathbf{n} \in \mathcal{M})$ are the sets of vector and time variables respectively where $z_e = (x_e, v_e)$, and

$$\begin{aligned} \mathcal{D} := & \{\mathbf{t}_{\mathcal{M}} = (t_{\mathbf{n}}) \in (\mathbb{R}^+)^{|\mathcal{M}|} : (\ell' - 1)\tau < t_{\mathbf{n}} < \ell'\tau \text{ if } \mathbf{n} \in \mathbb{M}_{\ell'}, \\ & t_{\mathbf{n}} < t_{\mathbf{n}^+} \text{ if there exists an edge between } \mathbf{n}^+ \text{ and } \mathbf{n}, \text{ with } \mathbf{n}^+ \text{ being the higher atom}\}. \end{aligned} \quad (2.17)$$

For each $\mathbf{n} \in \mathcal{M}$, we denoted by (e_1, e_2) the two bottom edges at $\mathbf{n}$, and (e'_1, e'_2) the two top edges at $\mathbf{n}$, such that e_1 and e'_1 are serial (cf. Definition 2.1) and e_2 and e'_2 are serial. The set $\mathcal{E}_{\text{end}}^-$ is the set of all bottom ends (corresponding to initial velocities of the particles). For any bottom end $e \in \mathcal{E}_{\text{end}}^-$, define $\ell[e] := \ell[\mathbf{p}]$ as in Section 2.1.6 where $\mathbf{p}$ is the particle line containing e ; but this notion is not used in the proof and should be ignored below.

Finally, the distribution $\Delta = \Delta(z_{e_1}, z_{e_2}, z_{e'_1}, z_{e'_2}, t_{\mathbf{n}})$ in (2.16) is defined as follows. If $\mathbf{n}$ is a C-atom, then

$$\begin{aligned} \Delta(z_{e_1}, z_{e_2}, z_{e'_1}, z_{e'_2}, t_{\mathbf{n}}) := & \delta(x_{e'_1} - x_{e_1} + t_{\mathbf{n}}(v_{e'_1} - v_{e_1})) \cdot \delta(x_{e'_2} - x_{e_2} + t_{\mathbf{n}}(v_{e'_2} - v_{e_2})) \\ & \times \delta(|x_{e_1} - x_{e_2} + t_{\mathbf{n}}(v_{e_1} - v_{e_2})| - \varepsilon) \cdot [(v_{e_1} - v_{e_2}) \cdot \omega]_- \\ & \times \delta(v_{e'_1} - v_{e_1} + [(v_{e_1} - v_{e_2}) \cdot \omega]) \cdot \delta(v_{e'_2} - v_{e_2} - [(v_{e_1} - v_{e_2}) \cdot \omega]), \end{aligned} \quad (2.18)$$

where $\omega := \varepsilon^{-1}(x_{e_1} - x_{e_2} + t_{\mathbf{n}}(v_{e_1} - v_{e_2}))$ is a unit vector, and $z_- = -\min(z, 0)$. If $\mathbf{n}$ is an O-atom, then

$$\begin{aligned} \Delta(z_{e_1}, z_{e_2}, z_{e'_1}, z_{e'_2}, t_{\mathbf{n}}) := & \delta(x_{e'_1} - x_{e_1} + t_{\mathbf{n}}(v_{e'_1} - v_{e_1})) \cdot \delta(x_{e'_2} - x_{e_2} + t_{\mathbf{n}}(v_{e'_2} - v_{e_2})) \\ & \times \delta(|x_{e_1} - x_{e_2} + t_{\mathbf{n}}(v_{e_1} - v_{e_2})| - \varepsilon) \cdot [(v_{e_1} - v_{e_2}) \cdot \omega]_- \\ & \times \delta(v_{e'_1} - v_{e_1}) \cdot \delta(v_{e'_2} - v_{e_2}). \end{aligned} \quad (2.19)$$

The proof of (KF2): We start from (2.15), which involves an integral in all variables except those associated with the root particles in $r(\mathbb{M}) = H$. When calculating $\|E_H\|_{L^1}$, we integrate in them also, which creates the same expression as in (2.15) but with integral in all $\mathbf{z}_{p(\mathbb{M})}$ variables. With the flow operator $\mathcal{S}_{\mathbb{M}}$ and indicator function $\mathbb{1}_{\mathbb{M}}$ in (2.15), it is easy to see that the integral part in (2.15) exactly represents the **probability that all collisions and overlaps in $\mathbb{M}$ do occur** (with the pre-factor in (2.15), this becomes a normalized probability).

On the other hand, the integral part of (2.16) represents the same probability, because it is an integral in all time, space and velocity variables associated with the collisions and overlaps, under the Dirac functions that dictate these collisions and overlaps, and the pre-factor in (2.16) provides the same normalization. More precisely:

- We have $|\mathcal{E}| - 2|\mathcal{M}| = |p(\mathbb{M})|$ by noting that the total number of edges is equal to the total number $2\mathcal{M}$ of bottom edges at all atoms plus the total number of top ends, which is $p(\mathbb{M})$; this matches the pre-factor in (2.15) in view of the factor $\varepsilon^{(d-1)|H|}$ in (KF2).
- The $t_{\mathbf{n}}$ is just the time of the collision or overlap $\mathbf{n}$. For each edge e which represents a linear piece of trajectory, the v_e is the velocity of this particle in this piece, and x_e is the position of the particle in this piece, transported backward to time 0 (so the actual

position of the particle at any collision/overlap $\mathbf{n}$ is in fact $x_e + t_{\mathbf{n}}v_e$). As such, the Dirac functions in (2.18)–(2.19) clearly dictate the collisions (via the elastic collision formula) and overlaps (which do not change position and velocity). When we incorporate all these Dirac functions and integrate in all relevant variables, we get the probability of all these collisions and overlaps happening, as desired.

The above justifies (KF2) on a formal level. The precise proof is done essentially by repeating the case of one single collision $\mathbf{n}$, i.e.

$$\begin{aligned} \int_{\mathbb{R}^{4d} \times I} \Delta(z_{e_1}, z_{e_2}, z_{e'_1}, z_{e'_2}, t_{\mathbf{n}}) \cdot Q(z_{e'_1}, z_{e'_2}, t_{\mathbf{n}}) dz_{e'_1} dz_{e'_2} dt_{\mathbf{n}} \\ = \mathbb{1}_{\text{col}}(z_{e_1}, z_{e_2}) \cdot Q(x_{\text{out}}^1 - t_{\text{col}} \cdot v_{\text{out}}^1, v_{\text{out}}^1, x_{\text{out}}^2 - t_{\text{col}} \cdot v_{\text{out}}^2, v_{\text{out}}^2, t_{\text{col}}) \end{aligned} \quad (2.20)$$

for any $Q \geq 0$, where $I = [0, \tau]$, $\mathbb{1}_{\text{col}}$ is the indicator function that a collision/overlap happens between two particles $\mathbf{p}_1$ and $\mathbf{p}_2$ with initial position and velocity (z_{e_1}, z_{e_2}) , at a collision time $t_{\text{col}} \in I$, and $z_{\text{out}}^j = (x_{\text{out}}^j, v_{\text{out}}^j)$ for $j \in \{1, 2\}$ are the position and velocity of the particles $\mathbf{p}_1$ and $\mathbf{p}_2$ right after the collision; note that z_{out}^j and t_{col} are both functions of (z_{e_1}, z_{e_2}) . The proof of (2.20) is by elementary calculations, see the discussion after equation (7.10) in [3].

3. PART B: REDUCTION TO THE ALGORITHM

From now on, equipped with (KF1)–(KF2), the remaining goal is to estimate the right hand side of (KF2), i.e. $\sum \mathcal{I}(\mathbb{M})$. Note that everything is now **completely expressed in terms of molecules $\mathbb{M}$ and associated notions** (see for example the definition of $\mathcal{I}(\mathbb{M})$ in (2.16)). As such, from now on we will **only use notions of molecules and forget everything in the picture of particles, collisions trajectories etc.** In particular:

- Instead of “collisions” and “overlaps” we will talk about C- and O-atoms; instead of the number of “recollisions” we will talk about the circuit rank (i.e. number of independent cycles) in the molecule $\mathbb{M}$.
- In Section 2 we have identified (physical) particles with (abstract) particle lines in molecules, and used the same notion $\mathbf{p}$ for both of them. For now on, we will not refer to any physical particles and $\mathbf{p}$ will only refer to the particle lines defined in Definition 2.1.
- The main combinatorial properties of $\mathbb{M}$ that will be used in the proof below are (a)–(d) stated at the end of Section 2.1.6, which are also completely stated in terms of molecular notions such as particle lines.

In this section, we show how the needed estimate on $\sum \mathcal{I}(\mathbb{M})$ can be reduced to a cutting algorithm on the molecule $\mathbb{M}$. To focus on the core concepts, we adopt the following simplifications:

Simplifications:

- (1) We shall assume that $|H|$ is small relative to the number of cycles ρ in the molecule $\mathbb{M}$, i.e. $|H| \ll \rho$. The case when $|H| \gtrsim \rho$ is easier as the expression of E_H comes with a factor of $\varepsilon^{(d-1)|H|}$ (see (KF2)).
- (2) We shall assume that each single layer $\mathbb{M}_{\ell'}$ of $\mathbb{M}$ is a tree (i.e. has no cycles). In the general case, we would need to include an extra term $\mathfrak{R}$ in the formula for ρ below, which is the total number of C-molecules with cycles in all layers (i.e. the “clusters with recollisions”, note that each of them contributes at most Γ “recollisions” by truncation). The case $\mathfrak{R} \gtrsim \rho$ is easy to treat, while the case $\mathfrak{R} \ll \rho$ is the same as $\mathfrak{R} = 0$ (with an additional step of layer refining, see Section 4.4).

3.1. Bound on the number of molecules and Key Formula (KF3). In this subsection, we prove (KF3). This is done in [3], Propositions 6.12.

Step 1: Recall the layers $\mathbb{M}_{\ell'}$ of $\mathbb{M}$, the sets $p(\mathbb{M}_{\ell'})$ and $r(\mathbb{M}_{\ell'})$ with $s_{\ell'} = |r(\mathbb{M}_{\ell'})|$ as in Section 2.1.6. The first step is to prove that

$$\rho \sim \max_{1 \leq \ell' \leq \ell} s_{\ell'},$$

with constants possibly depending on $\mathfrak{L}$, for the circuit rank (i.e. number of independent cycles) ρ .

The idea is as follows. The $\lesssim$ direction is easy, as each layer is a forest by itself, the number of bonds joining different layers is $O(\max s_{\ell'})$ (as each such bond must belong to a particle line that *crosses layers*, i.e. one in some $r(\mathbb{M}_{\ell'})$), and adding each bond increases ρ by at most 1. As for the $\gtrsim$ direction: choose ℓ^* such that $s_{\ell^*} = |r(\mathbb{M}_{\ell^*})|$ is maximal. By repeated applying (c) at the end of Section 2.1.6 (and using that $r(\mathbb{M}_{\ell^*}) \subset p(\mathbb{M}_{\ell^*+1})$ by (b)), we know that all particle lines in $r(\mathbb{M}_{\ell^*})$ are connected to a particle line in $r(\mathbb{M}_{\ell}) = H$ in *layers $\ell^* + 1$ and higher*; in particular they can be divided into at most $|H| \ll \rho$ groups with each group being *connected to each other* in layers $\ell^* + 1$ and higher (this is most intuitive when (say) $|H|$ is a small number like 1 or 2, see Figure 7; but the general case $|H| \ll \rho$ is the same). On the other hand, by repeating (d) (and repeating (c) if needed), we see that each particle line in $r(\mathbb{M}_{\ell^*})$ must be connected to another particle line in $r(\mathbb{M}_{\ell^*})$ in *layers ℓ^* and lower*. As shown in Figure 7, this leads to $\gtrsim |r(\mathbb{M}_{\ell^*})| = s_{\ell^*}$ independent cycles, hence the $\gtrsim$ result. This proof is contained in [3], Lemma 6.11 (in our simplified case we have $\mathfrak{R} = |H'| = 0$), which is a simple counting argument⁷.

Step 2: Now, let the number of atoms in $\mathbb{M}$ be $m = |\mathbb{M}|$ which is bounded by $|\log \varepsilon|^C$ due to truncation, and the cyclic rank be ρ as above, then the number of such molecules is bounded by $C^m |\log \varepsilon|^{C\rho}$ (the first C does not depend on $\mathfrak{L}$ while the second C does, but we shall ignore such minor points below). This is proved by simply following the construction of $\mathbb{M}$: each $\mathbb{M}_{\ell'}$ is a forest (where the degree of each atom is ≤ 4), and it is well-known that there are only exponentially many such forests, leading to the C^m factor. Then, as in Section 2.1.6, we join the $\mathbb{M}_{\ell'}$ together to form $\mathbb{M}$; but this involves choosing and matching at most $\sum |r(\mathbb{M}_{\ell'})| = \sum s_{\ell'} \leq C\rho$ top/bottom ends (recall Figure 5), leading to at most $m^{C\rho} \leq |\log \varepsilon|^{C\rho}$ choices. This proves (KF3).

3.2. Cutting and Key Formula (KF4). In this subsection we introduce the operation of **cutting** and prove (KF4). We shall explain these for the simple molecule in Figure 8; there is not much extra that is needed to handle the general case since the operations are mostly local.

We shall make the simplification that $\mathbb{M}$ only contains C-atoms. The notion of cutting has to be modified slightly to deal with the presence of O-atoms (and O-atom chains). The reason is that fixing the value z_e for an edge e at an O-atom automatically fixes the value $z_{e'}$ for the edge e' serial to it (see (2.16), especially the Dirac functions in (2.19)), so they somehow need to be treated together. To deal with this complication, one has to introduce the notion of **ov-segments** etc., and modify the definition of cutting slightly in the presence of O-atoms. We don't go into this here, but see the discussions in Section 4.4.

We now define the **cutting** operation, in the absence of O-atoms, as follows: let $\mathbb{M}$ be a molecule and $A \subseteq \mathbb{M}$ be a set of atoms. By **cutting A as free**, we simply turn each bond between $\mathbf{n} \in A$ and $\mathbf{n}' \in \mathbb{M} \setminus A$ into a **free end** at $\mathbf{n}$ and a **fixed end** at $\mathbf{n}'$. See Figure 8 for an illustration: for the molecule in Figure 8, we cut it into two smaller molecules $\mathbb{M}_1$ and $\mathbb{M}_2$ where $\mathbb{M}_1$ consists of a single atom, and $\mathbb{M}_2$ consists of two atoms.

⁷Note however that the proof in [3] is written in the language of interaction diagrams which we are avoiding here, and needs to be translated into the molecule language. This will be done in the revision.

Using the cutting operation defined above, we can state (KF4) precisely as follows: if $\mathbb{M}$ is cut into $\mathbb{M}_1$ and $\mathbb{M}_2$, then

$$\mathcal{I}(\mathbb{M}) \leq \mathcal{I}(\mathbb{M}_1) \cdot \sup_{z_e: e \text{ fixed}} \mathcal{I}(\mathbb{M}_2), \quad (3.1)$$

where for molecules $\mathbb{M}_2$ with fixed ends, we define $\mathcal{I}(\mathbb{M}_2)$ as in (2.16) but **without integrating in z_e for fixed ends e and not counting fixed ends in $|\mathcal{E}|$** (the reason will be clear below).

To prove (3.1), first note that the pre-factor in (2.16) is not affected by cutting ($|\mathcal{M}|$ does not change, and $|\mathcal{E}|$ *without counting fixed ends* does not change because each bond is turned into one free end plus one fixed end). We then only need to consider the integral part, which is just a simple application of Fubini. Note that if a bond e is cut into a free end e_1 and a fixed end e_2 , we should assign the variables as $z_e = z_{e_1} = z_{e_2}$; now by Fubini we have

$$\begin{aligned} & \int_{\mathbb{R}^{2d|\mathcal{E}|} \times \mathbb{R}^{|\mathcal{M}|}} \prod_{n \in \mathcal{M}} \Delta(z_{e_1}, z_{e_2}, z_{e'_1}, z_{e'_2}, t_n) dz_{\mathcal{E}} dt_{\mathcal{M}} \\ &= \int_{\mathbb{R}^{2d|\mathcal{E}_1|} \times \mathbb{R}^{|\mathcal{M}_1|}} \prod_{n \in \mathcal{M}_1} \Delta(z_{e_1}, z_{e_2}, z_{e'_1}, z_{e'_2}, t_n) dz_{\mathcal{E}_1} dt_{\mathcal{M}_1} \\ & \quad \times \int_{\mathbb{R}^{2d|\mathcal{E}_2|} \times \mathbb{R}^{|\mathcal{M}_2|}} \prod_{n \in \mathcal{M}_2} \Delta(z_{e_1}, z_{e_2}, z_{e'_1}, z_{e'_2}, t_n) dz_{\mathcal{E}_2} dt_{\mathcal{M}_2}. \end{aligned}$$

Here $(\mathcal{E}_j, \mathcal{M}_j)$ are associated with $\mathbb{M}_j$ but not counting fixed ends, and we have replaced the $f^{\mathcal{A}}$ factors in (2.16) by 1 for simplicity, as they do behave as such once the values of x_e and v_e are bounded (an easy reduction⁸). Note also that $\mathcal{E}_2$ does not contain the fixed ends in $\mathbb{M}_2$; in the interior integral above, the values of z_e for **fixed ends e of $\mathbb{M}_2$** have been **fixed** as the values of the corresponding **free ends** in $\mathcal{E}_1$. By taking supremum in these values, we then get

$$\begin{aligned} & \int_{\mathbb{R}^{2d|\mathcal{E}_1|} \times \mathbb{R}^{|\mathcal{M}_1|}} \prod_{n \in \mathcal{M}_1} \Delta(z_{e_1}, z_{e_2}, z_{e'_1}, z_{e'_2}, t_n) dz_{\mathcal{E}_1} dt_{\mathcal{M}_1} \\ & \quad \times \sup_{z_e: e \text{ fixed}} \int_{\mathbb{R}^{2d|\mathcal{E}_2|} \times \mathbb{R}^{|\mathcal{M}_2|}} \prod_{n \in \mathcal{M}_2} \Delta(z_{e_1}, z_{e_2}, z_{e'_1}, z_{e'_2}, t_n) dz_{\mathcal{E}_2} dt_{\mathcal{M}_2}, \end{aligned}$$

which is just (3.1) by definition.

3.3. Elementary molecules and reduction to algorithm. As described in Section 1.2, we now apply a sequence of cutting operations to cut $\mathbb{M}$ into finitely many **elementary molecules**, which refers to 4 types of molecules: the $\{2\}$, $\{3\}$, $\{4\}$ and $\{33\}$ molecules, depicted in [Figure 9](#). Using (KF4) and consequently (1.2), we then need to estimate the values of $\mathcal{I}(\mathbb{M})$ for each elementary molecule, which is stated as follows.

- (1) **$\{2\}$ molecule:** this is a molecule with one degree 2 atom (here **degree** or **deg** is defined to be the number of bonds and free ends, **not counting fixed ends**; hence this atom will have 2 free ends and 2 fixed ends). We stipulate that the two free ends are always **both bottom** or **both top** ends, and never allow $\{2\}$ -molecules that do not satisfy this property (this will be important in the algorithm in Section 4 below). In this case, we have $\mathcal{I}(\mathbb{M}) \lesssim 1$ uniformly in (z_e) for fixed ends e .
- (2) **$\{3\}$ molecule:** this is a molecule with one deg 3 atom (1 fixed end and 3 free ends). This is the most ubiquitous molecule that appears in the cutting algorithm, for which we have $\mathcal{I}(\mathbb{M}) \lesssim \tau$ uniformly in the fixed ends.

⁸In reality, this reduction requires a somewhat delicate redistribution of the Gaussian weights in v carried by each copy of $f^{\mathcal{A}}$ in (2.16); see Section 9.2 of [3].

- (3) **{4} molecule**: this is a molecule with one deg 4 atom (no fixed ends). This is the main source of **loss**, for which we have $\mathcal{I}(\mathbb{M}) \lesssim \varepsilon^{-(d-1)}$.
- (4) **{33} molecule**: this is a molecule with two atoms each having deg 3 (i.e. the two molecules are connected by a bond, and each has 2 free ends and 1 fixed end in addition). This is the main source of **gain**, for which we have⁹ $\mathcal{I}(\mathbb{M}) \lesssim \varepsilon$ (or a positive power of ε).

4. PART C: THE ALGORITHM

Now, we start with a multi-layer molecule $\mathbb{M}$ that satisfies the key properties described in (a)–(d) at the end of Section 2.1.6, and apply a cutting sequence that cuts $\mathbb{M}$ into elementary molecules. By combining (1.2), the estimates in Section 3.3 and the proof in Section 1.3, it is clear that we only need to prove (KF5), which is

$$\#_{\{33\}} - 10d \cdot \#_{\{4\}} \geq c \cdot \rho, \quad (4.1)$$

for **some suitable cutting sequence**, where we recall $\rho \sim \max(s_{\ell'})$ as in Section 3.1. The proof of (4.1) requires several **cutting algorithms**, which will be the main topic of this section.

Before starting, let us first recall several useful facts about molecules and cutting:

- We often refer to $\mathbb{M}' \subseteq \mathbb{M}$ as a (sub-)molecule by regarding any bond between $\mathbb{M}'$ and $\mathbb{M} \setminus \mathbb{M}'$ as a *free end* in $\mathbb{M}'$. Also, once we cut out any elementary component S , then we will ignore it for the rest of the cutting algorithm and consider the molecule $\mathbb{M} \setminus S$ only.
- An atom $\mathbf{m}$ is a *parent* (resp. *child*) of another $\mathbf{n}$ if $\mathbf{m}$ is connected to $\mathbf{n}$ by a bond with $\mathbf{m}$ being above (resp. below) $\mathbf{n}$. We say, $\mathbf{m}$ is a *descendant* (resp. *ancestor*) of $\mathbf{n}$ if it is obtained by iteratively taking parents (resp. children) starting from $\mathbf{m}$. Note that this is a monotonic process, i.e. one cannot get back to the start by iteratively taking parents or iteratively taking children.
- If we cut any atom $\mathbf{n}$ as free, then this introduces a *bottom fixed end* at each *parent* of $\mathbf{n}$ (if exists), and a *top fixed end* at each *child* of $\mathbf{n}$ (if exists). Recall also that each atom has at most 2 parents and at most 2 children.

4.1. The two-layer case. It turns out that, most of the ideas in the cutting algorithm for general molecules $\mathbb{M}$, are already contained in a *simplified 2-layer case*, where we assume that

- (1) $\mathbb{M}$ has no O-atoms, and we ignore all the degeneracies (cf. Section 4.4);
- (2) $\mathbb{M}$ is a 2-layer molecule $\mathbb{M} = \mathbb{M}_U \cup \mathbb{M}_D$ with an “up layer” $\mathbb{M}_U$ and a “down layer” $\mathbb{M}_D$;
- (3) $\mathbb{M}_U$ and $\mathbb{M}_D$ are trees without fixed end, and no atom in $\mathbb{M}_D$ is parent of an atom in $\mathbb{M}_U$;
- (4) The number of atoms in $\mathbb{M}_U$ that are connected to an atom in $\mathbb{M}_D$ by a bond, is at least ρ .

The two-layer molecules can be viewed as those appearing in the calculation of the cumulants $E_H(2\tau)$, the first nontrivial case beyond the short-time regime. Moreover, the treatment of the general ℓ -layer case is not fundamentally harder than the 2-layer case, and only requires one extra step of *layer selection*, see Section 4.4. In Figure 10, we list some example of two-layer molecules that will feature in our discussions below.

4.2. The trichotomy. The treatment of the two-layer molecules defined in the previous section is really an interpolation between three extreme scenarios, leading to a key *trichotomy* in the proof. Each of the three scenarios will be represented by a toy model below, which is defined in terms of further properties of the layers $\mathbb{M}_U$ and $\mathbb{M}_D$, and each requires a different cutting strategy.

The classification into the three scenarios is based on the following two considerations:

⁹In reality, we need to consider the so-called **degeneracies** which complicates this picture; however they are relatively easy to deal with, see Section 4.4 for a brief discussion. We also ignore the distinction between **{33A}** and **{33B}** molecules which mostly occurs in the degenerate case.

- (1) Consider the set of atoms in $\mathbb{M}_U$ that are connected to *two* atoms in $\mathbb{M}_D$ by two bonds. Is this set big or small relative to ρ ?
- (2) Consider all the atoms in $\mathbb{M}_U$ that are connected to *at least one* atom in $\mathbb{M}_D$ by at least one bond. Are they *all connected* to each other (among themselves, without using the other atoms in $\mathbb{M}_U$)? Or are they *separated* by other atoms in $\mathbb{M}_U$?

The reason for considering these two questions will be clear later. For convenience, below we will refer to the atoms in (1) (those in $\mathbb{M}_U$ that are connected to *two* atoms in $\mathbb{M}_D$) as *2-connections*, and denote their number by $\#_{2\text{conn}}$. We also define the set in (2) (the atoms in $\mathbb{M}_U$ that are connected to *some* atom in $\mathbb{M}_D$) by X (note that $|X| \geq \rho$), and define the number of components of X (as defined in (2) above) by $\#_{\text{comp}}(X)$.

The four examples in Figure 10, which are examples of *toy models I–III* below, exactly correspond to different cases of two-layer molecules depending on different answers to the above questions. More precisely, we define a two-layer molecule to be:

- *Toy model I*: if $\#_{2\text{conn}} = 0$ and $X = \mathbb{M}_U$;
- *Toy model I plus*: if $\#_{2\text{conn}} = 0$ and $\#_{\text{comp}}(X) < (10d)^{-1} \cdot |X|$ (no 2-connections, X has few components);
- *Toy model II*: if $\#_{2\text{conn}} = 0$ and $\#_{\text{comp}}(X) \geq (10d)^{-1} \cdot |X|$ (no 2-connections, X has many components);
- *Toy model III*: if $\#_{2\text{conn}} \geq (10d)^{-1} \cdot |X|$ (many 2-connections).

We split the first toy model case into Toy model I and I plus just to simplify the exposition. The general case of a two-layer molecule is a relatively direct interpolation of the above toy models. In fact, the case when $0 < \#_{2\text{conn}} < (10d)^{-1} \cdot |X|$ is treated similar to toy models I–II but with extra step of removing 2-connections, see Section 4.4. As such, in what follows, we will focus on how to treat each toy model by itself.

4.3. Toy models I–III. We describe the strategy for each toy model. For more detailed description of the algorithm with proofs, see [4].

4.3.1. Toy model I. The idea for toy model I is to apply a naive algorithm in $\mathbb{M}_U$, where we *always* cut the lowest atom in $\mathbb{M}_U$, followed directly by a greedy algorithm in $\mathbb{M}_D$ to extract $\{33\}$ components whenever they appears in $\mathbb{M}_D$. The heuristic algorithm is described as follows and is depicted in Figure 11:

Definition 4.1 (Algorithm for Toy Model I). We define the following algorithm for toy models I.

- (1) At each time, choose a lowest atom $\mathbf{n}$ in $\mathbb{M}_U$, and cut $\mathbf{n}$ as free.
- (2) If this generates a pair of adjacent degree-3 atoms in $\mathbb{M}_D$, then we cut them as free, giving a $\{33\}$ elementary molecule (in what follows, we shall simply refer to this situation as the “appearance of a $\{33\}$ ” in $\mathbb{M}_D$). For technical reasons, we also need to include the case of a pair of degree-3 atoms in $\mathbb{M}_D$ separated by one degree-4 atom in $\mathbb{M}_D$, which we cut together as a $\{343\}$ molecules (that can be further cut into a $\{3\}$ molecule and a $\{33\}$ molecule), but this can be ignored for now. Repeat until no $\{33\}$ (or $\{343\}$) molecule exist in $\mathbb{M}_D$.
- (3) Go back to (1) and cut the next lowest atom in $\mathbb{M}_U$, and so on.
- (4) When all atoms in $\mathbb{M}_U$ have been cut, then cut $\mathbb{M}_D$ one atom at a time, such that: if any cutting causes the appearance of a $\{33\}$ (or $\{343\}$) molecule in $\mathbb{M}_D$ then go to (2) and remove them, then cut the next atom, and so on.

For the precise description of this algorithm, as well as the proof that it yields (4.1), see Section 1.1 of [4]. Here, we content ourselves with the illustration of this algorithm by an example as described in Figure 11, and making a few remarks about the intuition behind it:

(1) The reason we always cut a lowest atom in $\mathbb{M}_U$, is to guarantee that $\mathbb{M}_U$ does not contain any *top fixed end* (a top end that is fixed) in the whole process (this is because, cutting any *lowest* atom, which has no *child* in $\mathbb{M}_U$ at the time it is cut, will not generate any *top fixed end* in $\mathbb{M}_U$). This property, as well as similar ones, will play an important role below; we refer to them as *monotonicity* properties. The significance of the monotonicity property is to guarantee that there cannot be any non-elementary molecules produced in the process. In fact, the only non-elementary molecules of *one atom* are those of deg 1 or deg 2 with both top and bottom fixed ends, but they cannot be produced under the monotonicity assumption.

(2) A drawback of the naive algorithm of cutting lowest is that, if the lowest atom happens to have deg 4, then we will get a bad $\{4\}$ molecule. However, it can be shown that for toy model I, $\mathbb{M}_U$ can have *at most one lowest atom*¹⁰ (see Figure 12 and [4] for a proof). This allows us to control the loss of $\{4\}$ molecules.

(3) The overall idea is to (i) keep cutting lowest atoms in $\mathbb{M}_U$ to guarantee monotonicity, and thus guarantee that $\mathbb{M}_U$ does not produce any “forbidden” non-elementary molecule (we *do not* intend to gain from $\mathbb{M}_U$), and (ii) extract gain from $\mathbb{M}_D$ whenever possible. In the end all the bonds between $\mathbb{M}_U$ and $\mathbb{M}_D$ lead to gains, and all such gains are captured in $\mathbb{M}_D$.

4.3.2. *Toy model I plus.* Toy model I plus is viewed as an extension of toy model I, and the strategy is similar; in fact, the greedy algorithm in $\mathbb{M}_D$, i.e. step (2) in the algorithm for toy model I, is *exactly the same* for toy model I plus. The difference occurs in $\mathbb{M}_U$, as follows:

A key feature for toy model I is that $\mathbb{M}_U$ has at most one lowest atom, which reduced the number of bad $\{4\}$ molecules, see (2) in Section 4.3.1 above. Now for toy model I plus, the $\mathbb{M}_U$ may have many lowest atoms, so a naive “cutting lowest” algorithm will not work. Instead, we shall avoid deg 4 atoms and choose a *lowest deg 3* atom $\mathbf{n}$ each time. This leads to a number of changes:

(i) if a deg 3 atom $\mathbf{n}$ is not lowest (it is only lowest among deg 3 atoms, which means it may have children which are deg 4), then cutting it will create *top fixed ends* in $\mathbb{M}_U$. However, this top fixed end will only occur at the children (and subsequently descendants) of $\mathbf{n}$, which now has no *bottom fixed ends* (since these children nodes must have deg 4 before $\mathbf{n}$ is cut, by definition). Denote by $S_{\mathbf{n}}$ the set of its descendants (including $\mathbf{n}$).

(ii) Subsequently, we can cut *all descendants* of $\mathbf{n}$, going *from high to low*, and this maintains the property that *no bottom fixed end* occurs at any descendants of $\mathbf{n}$. Therefore, the monotonicity property in Section 4.3.1 (1) is now replaced by the following: there is no *top fixed end outside* the set $S_{\mathbf{n}} \setminus \{\mathbf{n}\}$ and no *bottom fixed end inside* the set $S_{\mathbf{n}} \setminus \{\mathbf{n}\}$, where $S_{\mathbf{n}}$ is the set of descendants of $\mathbf{n}$.

This idea leads to the following modification of the naive “cutting lowest” atom in $\mathbb{M}_U$ (i.e. Step (1) in the algorithm of Toy Model I), which is (essentially) the **UP** algorithm. Note that it preserves the new version of monotonicity described in (ii) above, and thus does not create “forbidden” non-elementary components. Moreover, it also minimizes the number of bad $\{4\}$ molecules.

Definition 4.2 (The **UP** Algorithm). We define the following **UP** algorithm.

- (1) Each time choose a lowest atom $\mathbf{n}$ among all deg 3 atoms (if there are no degree-3 atom, choose a lowest degree 4 atom), and let $S_{\mathbf{n}}$ be the set of descendants of $\mathbf{n}$.
- (2) Cut all atoms in $S_{\mathbf{n}}$ as free one by one, each time cutting a *highest* atom $\mathbf{m}$. If at any point, $\mathbf{m}$ has a parent $\mathbf{m}^+$ or a child of $\mathbf{m}^-$ that is also of degree 3, we cut $\{\mathbf{m}, \mathbf{m}^{\pm}\}$ together as a $\{33\}$ molecule.

¹⁰Here lowest atom means lowest before any cutting. Note that an atom $\mathbf{n}$ which is not initially lowest may become lowest during cutting; however, in this case some child of $\mathbf{n}$ must have been cut before, creating a fixed end at $\mathbf{n}$, so $\mathbf{n}$ cannot be deg 4 when it is cut.

- (3) Cut any possible deg 2 atoms until there is no more deg 2 atom, and go back to (1) and choose the next lowest deg 3 atom, and so on.

This algorithm can be applied to any molecule $\mathbb{M}$ without any top fixed ends. See Figure 13 for an illustration of how it works, and Figure 14 for an illustration of why it preserves monotonicity in $\mathbb{M}$. Similar to **UP**, we define the algorithm **DOWN** in exactly the same way, but with switching top and bottom, highest and lowest, descendants to parents, etc. In other words, **DOWN** is the algorithm **UP** applied to the inverted molecule of $\mathbb{M}$ obtained by turning the picture of $\mathbb{M}$ upside down.

Now, in the case of Toy Model I plus, we would like to combine the **UP** algorithm (instead of naive algorithm) in $\mathbb{M}_U$ with the greedy algorithm in $\mathbb{M}_D$. However, for the **UP** we need to restrict to the part of $\mathbb{M}_U$ that *does not affect* $\mathbb{M}_D$, i.e. to the set $\mathbb{M}_U \setminus X$ of atoms in $\mathbb{M}_U$ that are not adjacent to $\mathbb{M}_D$ (otherwise the operations in $\mathbb{M}_D$ may break the monotonicity in $\mathbb{M}_U$ by creating bottom fixed ends).

On the other hand, we still apply the naive strategy (i.e. Step (1) of Definition 4.1) within X , which may lead to bad $\{4\}$ molecules. However, the number of $\{4\}$ components is *comparable to* $\#_{\text{comp}}(X)$, because again *each component* of X can have *at most one lowest atom* (see Figure 12 or the proof in [4]). This loss is negligible, *under the assumption that X has few components in toy model I plus*.

In summary, the strategy for toy model I plus can be described as:

Definition 4.3 (Algorithm for Toy Model I plus). We define the algorithm for toy model I plus:

- (1) Choose a lowest atom $\mathbf{n}$ in $X \cup (\mathbb{M}_U)_3$ where $(\mathbb{M}_U)_3$ is the set of deg 3 atoms in $\mathbb{M}_U$;
- (2) Then, we apply the step (2) in **UP** to cut all atoms in $S_{\mathbf{n}}$ (i.e. set of descendants of $\mathbf{n}$ in $\mathbb{M}_U$), but at any time, if $\{33\}$ (or $\{343\}$) molecule occurs in $\mathbb{M}_D$, then we immediately repeat step (2) in Definition 4.1 to remove them.
- (3) When all atoms in $\mathbb{M}_U$ have been cut, we repeat step (4) of Definition 4.1 to finish off.

This algorithm is illustrated in Figures 15–16. For more detailed description of the algorithm and a proof that it yields the target estimate (4.1), see Section 1.2 of [4]. In this algorithm, by choosing the lowest atom in $X \cup (\mathbb{M}_U)_3$ in (1), we are applying the *naive* “choosing lowest” strategy in X and the **UP** “choosing lowest deg 3” strategy in $\mathbb{M}_U \setminus X$, as in the intuitive discussion above.

4.3.3. Toy model II. Now consider toy model II. Here we need a completely different strategy. It is based on the following observation: if we cut $\mathbb{M}_D$ as free (and subsequently cut it using **DOWN** which is the dual of **UP**), then the resulting $\mathbb{M}_U$ contains no top fixed end and no deg 2 atom, and exactly those atom in X have deg 3.

Here, the important picture is that of a molecule $\mathbb{M}_U$ with only deg 3 and 4 atoms, and no top fixed ends. Let $(\mathbb{M}_U)_3 = X$ be the set of deg 3 atoms, then the assumption $\#_{\text{comp}}(X) \geq (10d)^{-1} \cdot |X|$ in toy model II implies that X has many components, i.e. *the deg 3 atoms are separated by deg 4 atoms*, see Figure 17. In this case, one can apply a variant of **UP** algorithm, called the **3COMPUP** algorithm, which makes clever (instead of arbitrary) choices for the lowest deg 3 atoms in the process. This allows to obtain *one $\{33\}$ molecule per connected component of X* , which leads to enough $\{33\}$ molecules under the many components assumption of X .

Essentially, instead of choosing one lowest deg 3 atom in **UP**, we choose *two* lowest deg 3 atoms at one time, and cut them and all their descendants, to guarantee the presence of one $\{33\}$ molecule in this process, see Figure 17. It can be described as follows. First cut $\mathbb{M}_D$ as free. Then:

- (1) Each time, choose a connected, no-bottom subset $A \subseteq \mathbb{M}_U$ (i.e. the child of any atom in A is again in A) with exactly two deg 3 atoms. Such set can be shown to always exist.

- (2) Apply the **UP** algorithm within A to cut all atoms in A (cutting a $\{33\}$ molecule if possible).
 The go back to (2) and choose the next set A , and so on.

For the precise definition of this algorithm and the proof that yields the desired bound (4.1), see Section 1.3 in [4].

4.3.4. Toy model III. Finally, for toy model III which contains many 2-connections, we apply a simple variant of **DOWN**, called **2CONNDN**, to gain $\{33\}$ molecules from each of these 2-connection atoms. This is based on the following observation: let $\mathbf{n} \in \mathbb{M}_U$ be a 2-connection which is connected to $\mathbf{m}_1, \mathbf{m}_2 \in \mathbb{M}_D$. Suppose we apply **DOWN** in $\mathbb{M}_D$, in which we cut atoms in $\mathbb{M}_D$ one by one; assume $\mathbf{m}_1$ is cut before $\mathbf{m}_2$, then the rules of **DOWN** (dual to **UP**) implies that $\mathbf{m}_2$ will have deg 3 when it is cut, while at the mean time $\mathbf{n}$ also has deg 3 (because $\mathbf{m}_1$ has been cut). This then leads to a $\{33\}$ molecule for each $\mathbf{n}$, which we then exploit to gain enough $\{33\}$ molecules using that $\#_{2\text{conn}} \geq (10d)^{-1} \cdot |X|$. An example of this algorithm is given in Figure 18 and see Definition 1.20 in [4] for precise version:

- (1) Apply **DOWN** in $\mathbb{M}_D$, but whenever an atom $\mathbf{n} \in \mathbb{M}_D$ forms a $\{33\}$ molecule with an atom $\mathbf{m} \in \mathbb{M}_U$, cut it immediately, then proceed with **DOWN** in $\mathbb{M}_D$.

4.4. Reduction to toy models. Finally, we briefly discuss the extra pieces that need to be added to the above toy models, in order to treat the general case:

The O-atoms. In reality, the molecule $\mathbb{M}$ may contain O-atoms. This complicates the definition of cutting, when the role of bonds is generally replaced by those of *ov-segments*. The latter are a chain of O-atoms separating two C-atoms in a given particle line. We then need to translate the relevant notions, such as parent/child and components, into ov-parent/ov-child and ov-components etc., associated with the ov-segments. This translation, while tedious, *will not cause any essential change* to the proof.

Layer selection. To reduce from multi-layer to 2-layer case, we need to perform one key step of *layer selection*. Start with a layer ℓ_U with $s_{\ell_U} = |r(\mathbb{M}_{\ell_U})|$ being largest. By the proof in Section 3.1, all the particle lines in $r(\mathbb{M}_{\ell_U})$ are connected to each other via layers between 1 and ℓ_U ; then choose the *highest* layer ℓ_D , such that *sufficiently many* of these particle lines are connected to each other via layers between ℓ_D and ℓ_U . In this way, most recollisions can be located to happen *at layer* ℓ_D , so we can simply *isolate layers* ℓ_U and ℓ_D (with certain sets between ℓ_U and ℓ_D attached to ℓ_U) which already contain enough information, to reduce to the 2-layer case.

Layer refining. In the above, we have assumed each single layer of $\mathbb{M}$ is a tree/forest. In reality, if some layer contains a cycle, we can divide this layer into two *thin-layers*, such that each of them has no cycle. In general, we can divide each layer of $\mathbb{M}$ into at most $(\Gamma + 1)$ thin layers, where Γ is the recollision truncation threshold, such that each cluster in each layer contains at most Γ cycles/recollisions (this can be stated purely in terms of molecules). This then allows us to perform the same layer selection and toy model analysis as above.

Degeneracies. The degeneracies occur when the vector $z_e = (x_e, v_e)$ for certain edges e , which may appear as fixed ends of $\{33\}$ molecules, satisfy certain extra restrictions, which affects the gain from $\{33\}$ molecules. To deal with such degeneracies, our strategy is as follows: if there are few degeneracies (compared to ρ), then they are negligible and we just proceed with the usual algorithm; if there many of them, we perform a much simpler *alternative cutting sequence* where most atoms are cut as $\{3\}$ molecules, while the many *extra restrictions* will lead to extra gains in integration, which will be sufficient.

Pre-processing. Finally, the cutting process associated with layer selection may lead to fixed ends in the resulting 2-layer molecule; similarly, if $0 < \#_{2\text{conn}} \ll |X|$ for the 2-layer model, we also need to first cut these 2-connections which introduces fixed ends. To deal with these fixed ends, we need

an extra *pre-processing* step to first get rid of any $\{33\}$ (and $\{343\}$ molecules) in $\mathbb{M}_D$. This is a straightforward process, see [3], Definition 11.8–Proposition 11.9 for more details.

5. SOME CONCLUDING REMARKS

We end these notes with two remarks concerning Sections 12–13 in [3].

The error in (KF1) and the truncation of the dynamics. The truncation of the dynamics capping the number of particles by Λ_ℓ and number of recollisions by Γ for each cluster in each time layer, introduces an error $f_s^{\text{err}}(\ell\tau)$ in (2.1). This error is generated when the original flow deviates, at some point, from that of the truncated dynamics, by having a cluster with either $\geq \Lambda_\ell$ particles or $\geq \Gamma$ recollisions. The idea to treat this error term is to consider the *first layer* $[(\ell' - 1)\tau, \ell'\tau]$ where this deviation happens; in particular the s -particle marginals at time before $(\ell' - 1)\tau$ is well-controlled by the same proof. Then, in the layer $[(\ell' - 1)\tau, \ell'\tau]$, we truncate the dynamics until *the first time* where the deviation occurs, which produces a molecule with a cluster either of size between Λ_ℓ and $2\Lambda_\ell$ (which is easy to treat), or with the number of recollisions between Γ and 2Γ .

The analysis of the layers $\ell' - 1$ and below goes exactly like above; we only need to consider the layer ℓ' , which essentially reduces to proving that

- If a cluster contains $\geq \Gamma$ cycles/recollisions, then we gain¹¹ $\varepsilon^{q(\Gamma)}$ with $q(\Gamma) \rightarrow \infty$ as $\Gamma \rightarrow \infty$.

The main enemy in the above statement is when a small group of particles within distance $O(\varepsilon)$ of each other produce many recollisions within time $O(\varepsilon)$. To handle this, we need *a new ingredient*, namely the absolute upper bound on the number of collisions given the number of particles, which is proved by Burago-Ferleger-Kononenko. Basically, if Γ is sufficiently large, then these $\Gamma + 1$ recollisions must involve sufficiently many particles, with each new particle providing a $\{33\}$ -molecule in a suitable algorithm. The latter algorithm is inspired by that in Sections 10–11 in [3] to extract the needed gain from this single cluster with many recollisions. See [3], Section 13.

The f^A analysis in [3], Section 12. As explained in (2.13), the expansion of the f^A is in terms of molecules with only one root particle. Such molecules might not be trees, but if they are not, then one can apply some basic algorithms (namely the **UP** algorithm) in order to prove that they are lower order. As such, one can reduce to the case when all the molecules in (2.13) are trees.

Furthermore, a cancellation first observed in [1], shows that molecules that are not *binary trees* cancel completely. The contribution of the remaining leading order molecules to (2.13), given by binary trees, can be mapped directly order-by-order to the time τ iterates of the Boltzmann equation with initial data given by $f^A((\ell - 1)\tau)$. By induction, this allows us to approximate $f^A(\ell'\tau)$ by $n(\ell'\tau)$ for all $1 \leq \ell' \leq \ell$, where $n(t)$ is the solution of the Boltzmann equation.

¹¹In [3], due to technical reasons, we can only achieve $q(\Gamma) \geq d + 1$ as $\Gamma \rightarrow \infty$, but this is already enough as we only need to gain one factor of $N^{-1} \sim \varepsilon^{d-1}$.

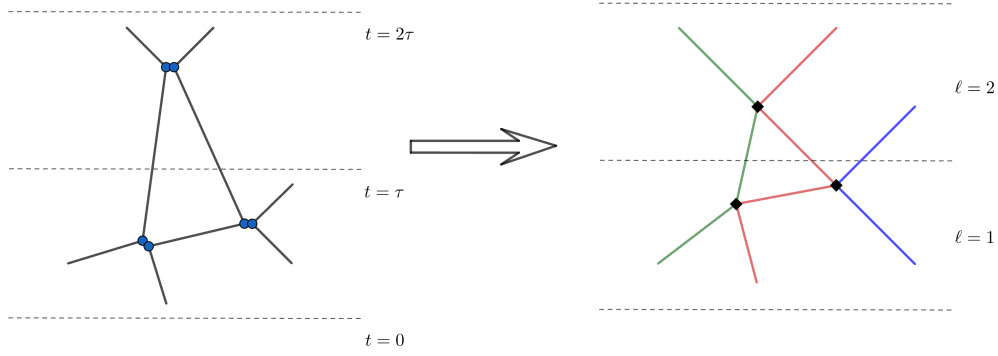


FIGURE 1. Converting collision trajectory to C-molecule. Each collision is converted to a vertex called atom, and the linear motion between two collision is represented by a bond. The trajectory of each single particle is mapped to the *particle line* corresponding to the particle, which is depicted in different colors in the figure on the right. The part of each particle before the first collision and after the last collision is given by the two free ends (one top and one bottom) for the particle line.

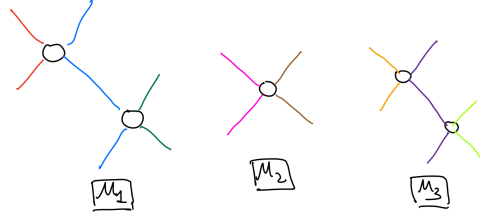


FIGURE 2. A collection of three C-molecules $\mathbb{M}_1$, $\mathbb{M}_2$, and $\mathbb{M}_3$, which correspond to three clusters for the N -particle dynamics. The C-atoms are represented by a hollow circle.

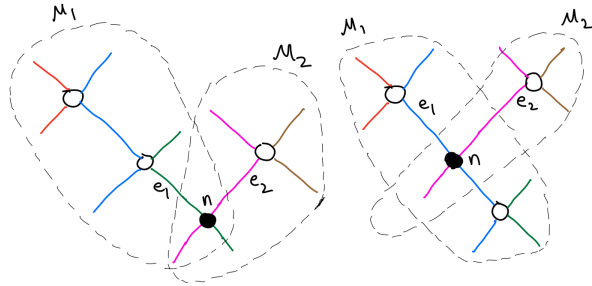


FIGURE 3. Two examples of the molecules $\mathbb{M}$ that result from joining the two C-molecules $\mathbb{M}_1$ and $\mathbb{M}_2$ from Figure 2 via an O-atom (depicted by the solid black dot). In the first example, the green particle line from $\mathbb{M}_1$ overlaps with the magenta particle line from $\mathbb{M}_2$ before it undergoes its collision with the blue particle line in $\mathbb{M}_1$. In the second example, the blue particle line from $\mathbb{M}_1$ overlaps with the magenta particle line from $\mathbb{M}_2$ in between the two collisions that the blue particle line undergoes in $\mathbb{M}_1$.

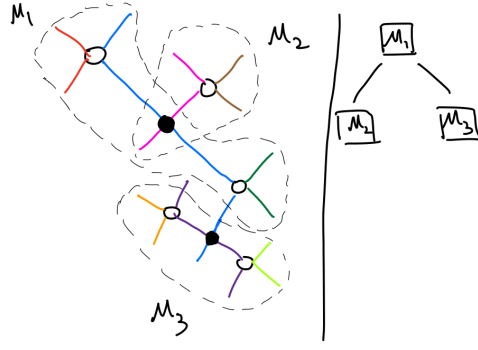


FIGURE 4. An example of the molecule $\mathbb{M}$ formed by joining clusters/C-molecules by O-atoms (as the result of applying the Penrose argument) on the left side, and the corresponding cluster graph $F = G_{\mathbb{M}}$ (which is a forest) on the right side. Here F is just one tree with three nodes given by three clusters $\mathbb{M}_1, \mathbb{M}_2, \mathbb{M}_3$. We use the same C-molecules from Figure 2 as $\mathbb{M}_1, \mathbb{M}_2, \mathbb{M}_3$. These C-molecules can be recovered from the full molecule $\mathbb{M}$ above, by deleting the O-atoms and joining any two serial edges at that atom. This is shown in the figure by the dotted selections.

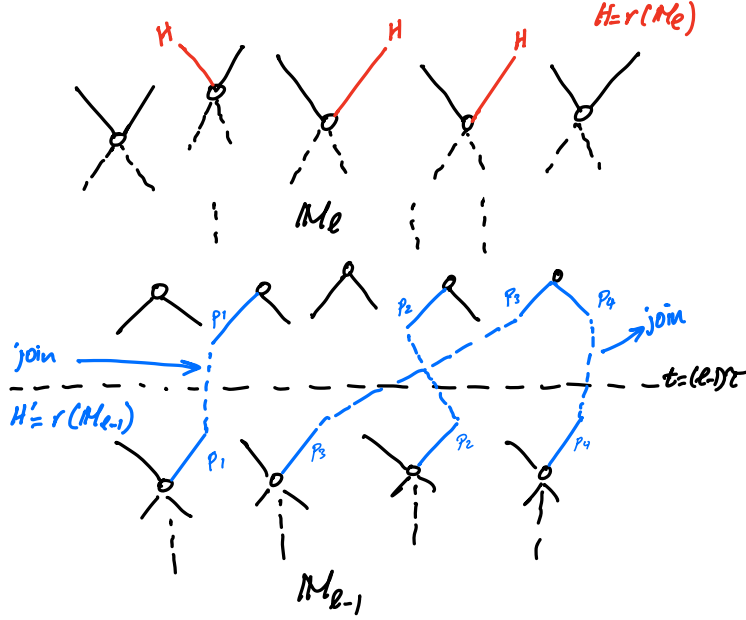


FIGURE 5. The method by which the layer ℓ molecule $\mathbb{M}_\ell$ is joined to the lower layer molecule $\mathbb{M}'$. $\mathbb{M}_\ell$ has bottom ends corresponding to particles in H' , and $\mathbb{M}'$ has its root particles $r(\mathbb{M}') = H'$. We simply join the bottom ends in $\mathbb{M}_\ell$ to the top ends in $\mathbb{M}'$ that correspond to the same particle in H' to form a bond going from a layer ℓ atom to an atom with a layer $\leq \ell - 1$. **Here and below**, when it is not necessary to distinguish all different particle lines, we will not use a different color for each particle line, and save the colors for other purposes (here for the particle lines in H and H').

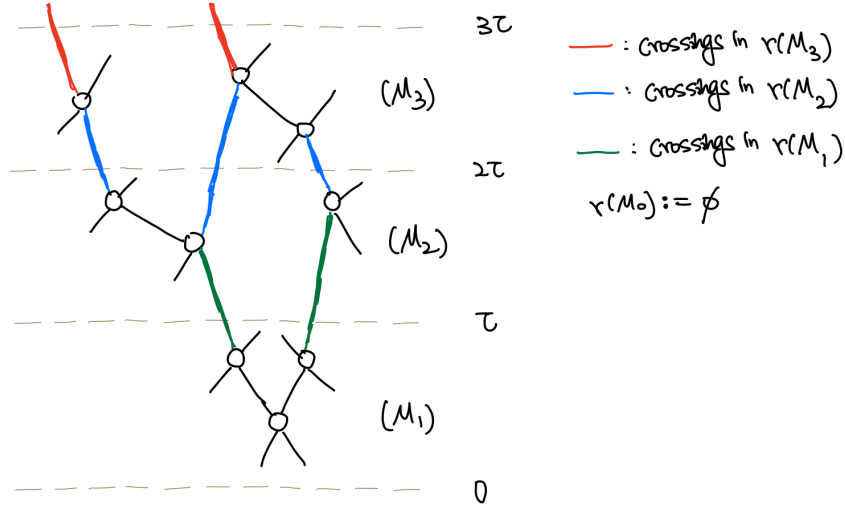


FIGURE 6. An example of the 3-layer molecule $\mathbb{M}$ satisfying properties (a)–(d) in Section 2.1.6. The red edges belong to those particle lines in $r(\mathbb{M}_3) = H$, and intuitively represent that they *cross into* layer $\mathbb{M}_3$. Same for blue and green edges (corresponding to $r(\mathbb{M}_2)$ and $r(\mathbb{M}_1)$; note that different $r(\mathbb{M}_j)$ may have common particle lines). The $p(\mathbb{M}_j)$ sets consist of $r(\mathbb{M}_j)$ and those particle lines containing the **top ends** in $\mathbb{M}_j$. Think in this way: the 2 red particles ($r(\mathbb{M}_3)$) are roots. Their dynamics in $\mathbb{M}_3$ leads to 5 total particles ($p(\mathbb{M}_3)$), 3 of which ($r(\mathbb{M}_2)$ in blue, which may partly coincide with red) cross into $\mathbb{M}_2$. Their dynamics in $\mathbb{M}_2$ leads to 5 total particles ($p(\mathbb{M}_2)$), 2 of which ($r(\mathbb{M}_1)$ in green) cross into $\mathbb{M}_1$, and so on. It is easy to verify (a)–(d) in Section 2.1.6; moreover **any crossing particle line is connected to another crossing particle line in layers below**.

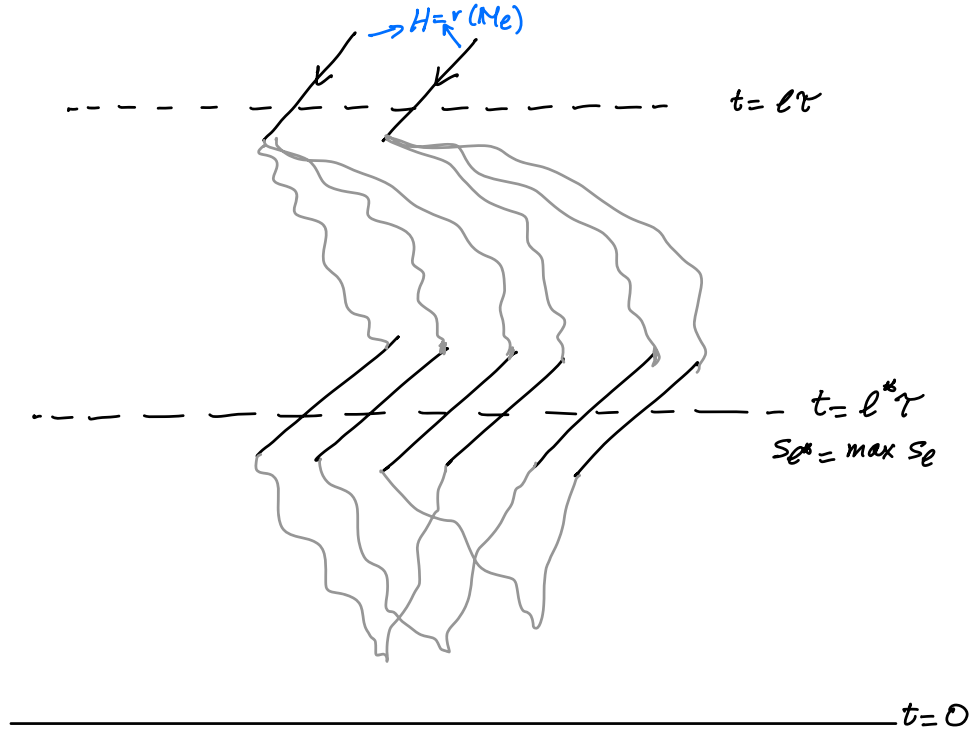


FIGURE 7. Here, we suppose that ℓ^* is the layer with maximal s_{ℓ^*} , and that $|H| = 2$. Each of the s_{ℓ^*} particle lines crossing into layer ℓ^* is connected to another such particle line in lower layers. Also, all those particle lines are connected to H in the layers above ℓ^* . This creates ρ cycles where $\rho \sim s_{\ell^*}$.

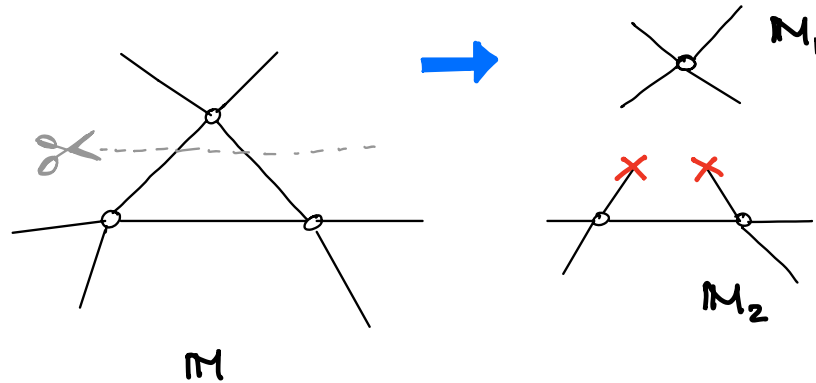


FIGURE 8. The cutting process: A molecule $\mathbb{M}$ is cut at two bonds to produce two molecules $\mathbb{M}_1$ and $\mathbb{M}_2$. Each bond that is cut in $\mathbb{M}$ becomes two ends, which are a free end in $\mathbb{M}_1$ and a fixed end in $\mathbb{M}_2$ (represented by a red cross). In this case we say that $\mathbb{M}_1$ is cut as free from $\mathbb{M}$ (and $\mathbb{M}_2$ is cut as fixed).

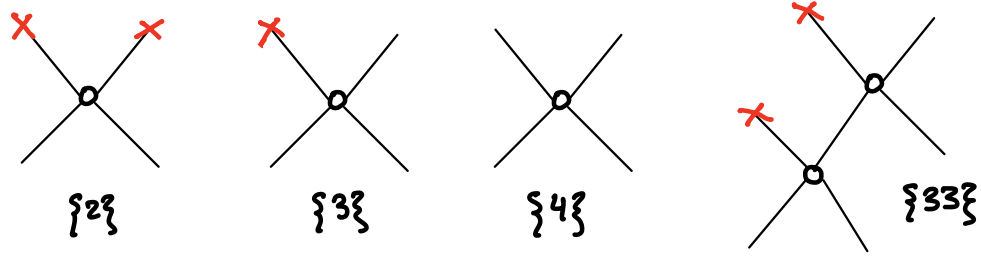


FIGURE 9. Elementary molecules. Fixed ends are denoted by a red cross. The first picture is a $\{2\}$ -molecule which has two fixed ends (either both on top or both on bottom). The second picture is a $\{3\}$ -molecule which has one fixed end (that can be at any of the ends). The third picture is a $\{4\}$ -molecule having no fixed ends. The fourth picture is a $\{33\}$ -molecule which has two connected atoms, each having exactly one fixed end.

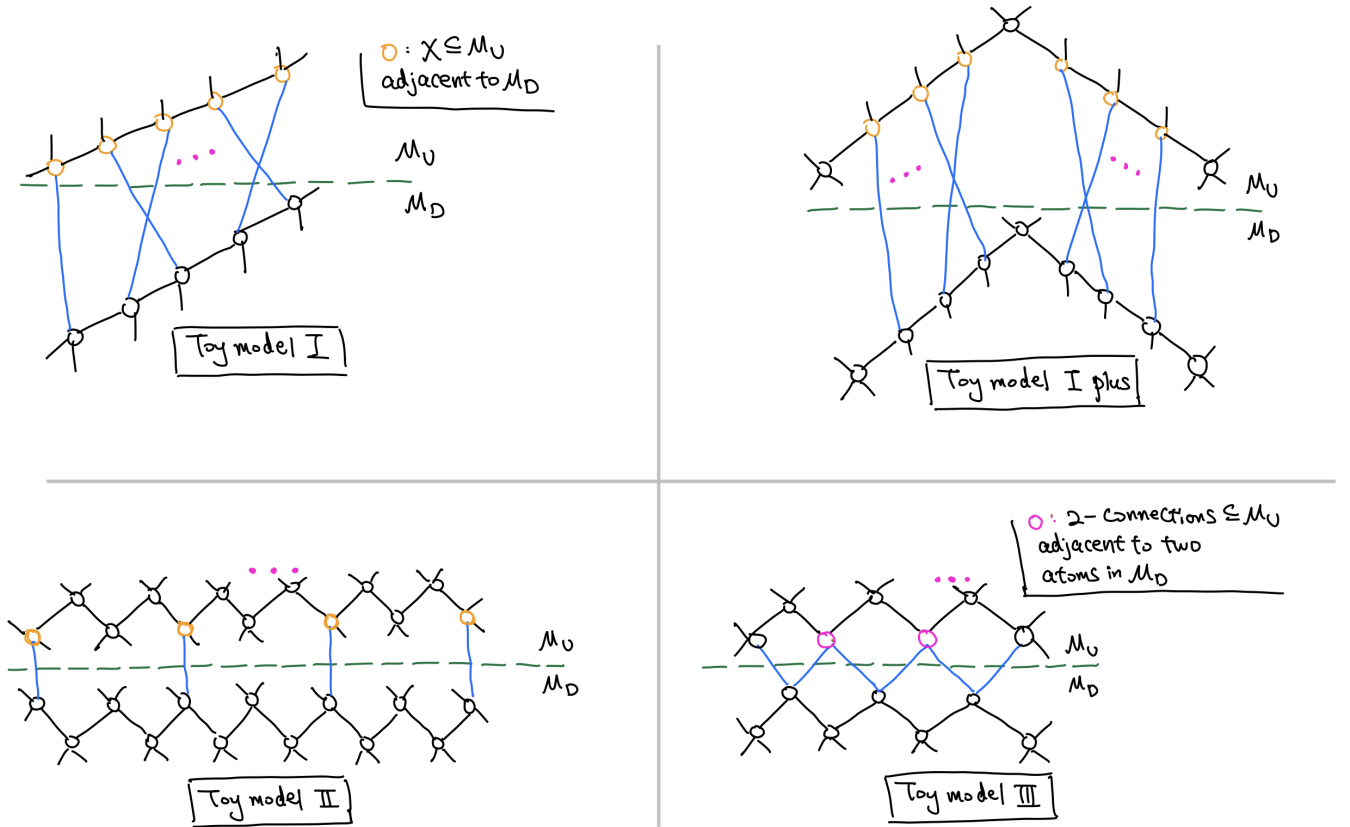


FIGURE 10. Examples of toy models. Upper left: toy model I ($X = \mathbb{M}_U$). Upper right: toy model I plus (where $\#_{\text{comp}}(X) = 2$). Lower left: toy model II (where $\#_{\text{comp}}(X) = |X|$). Lower right: toy model III (where $\#_{2\text{conn}} \sim |X|/2$). The blue bonds connect $\mathbb{M}_U$ and $\mathbb{M}_D$. The orange and pink atoms indicate those in X and 2-connections respectively. The pink dots represent patterns that can be repeated.

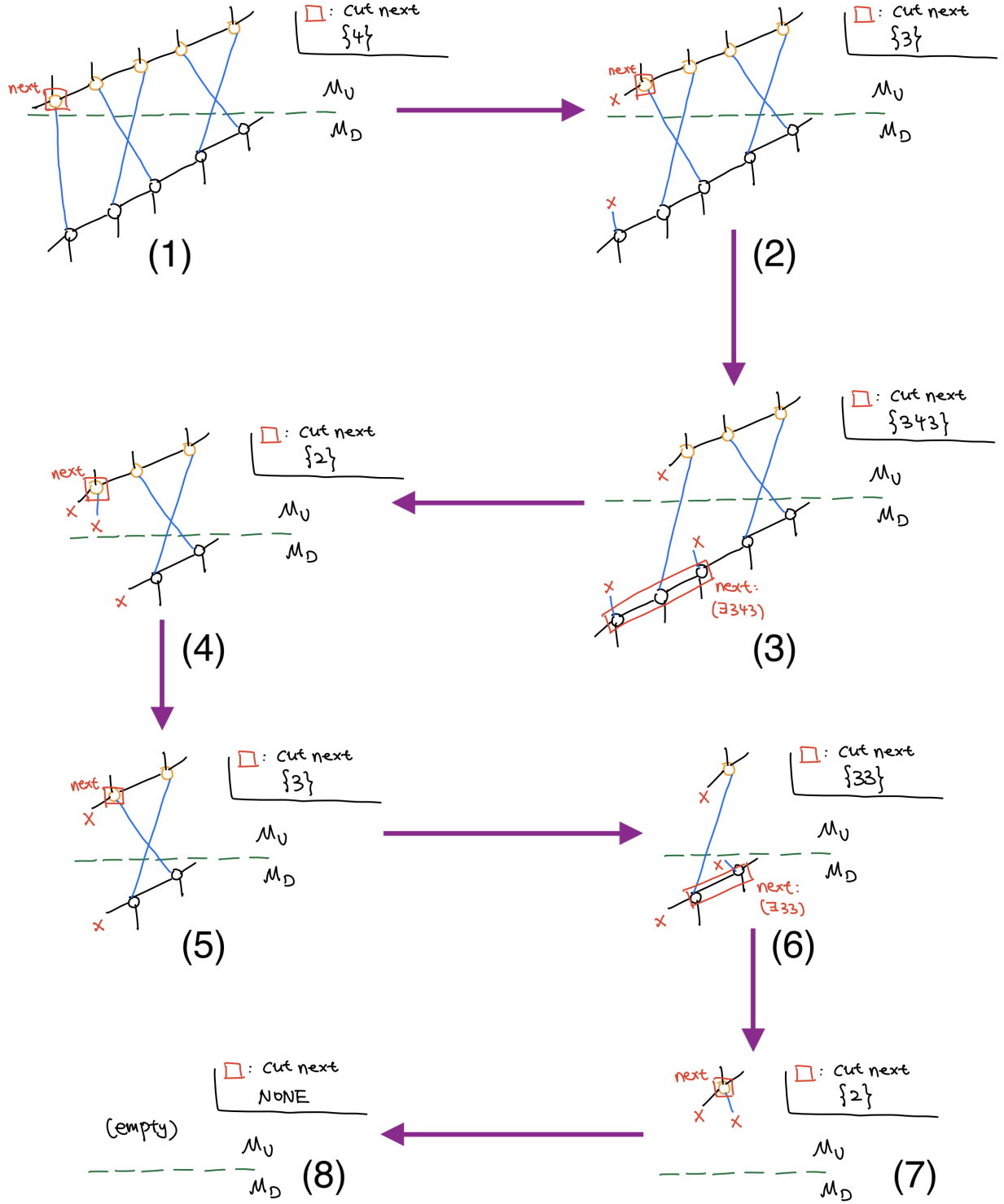


FIGURE 11. Algorithm for toy model I, applied to the example in Figure 10. Following the rule of the algorithm (Definition 4.1), each time we cut the atom(s) in the red box, creating fixed ends. The type of the elementary molecules to be cut is listed at the upper right corner.

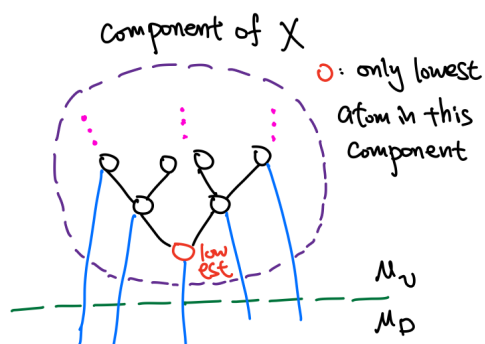


FIGURE 12. Illustration of each component of X (or M_U if $X = M_U$ in toymodel I plus). It has a unique lowest atom (in red). The other atoms can only be constructed from the red one by taking *parents*, as they must have a bond connecting to M_D (in blue) and cannot have a third child.

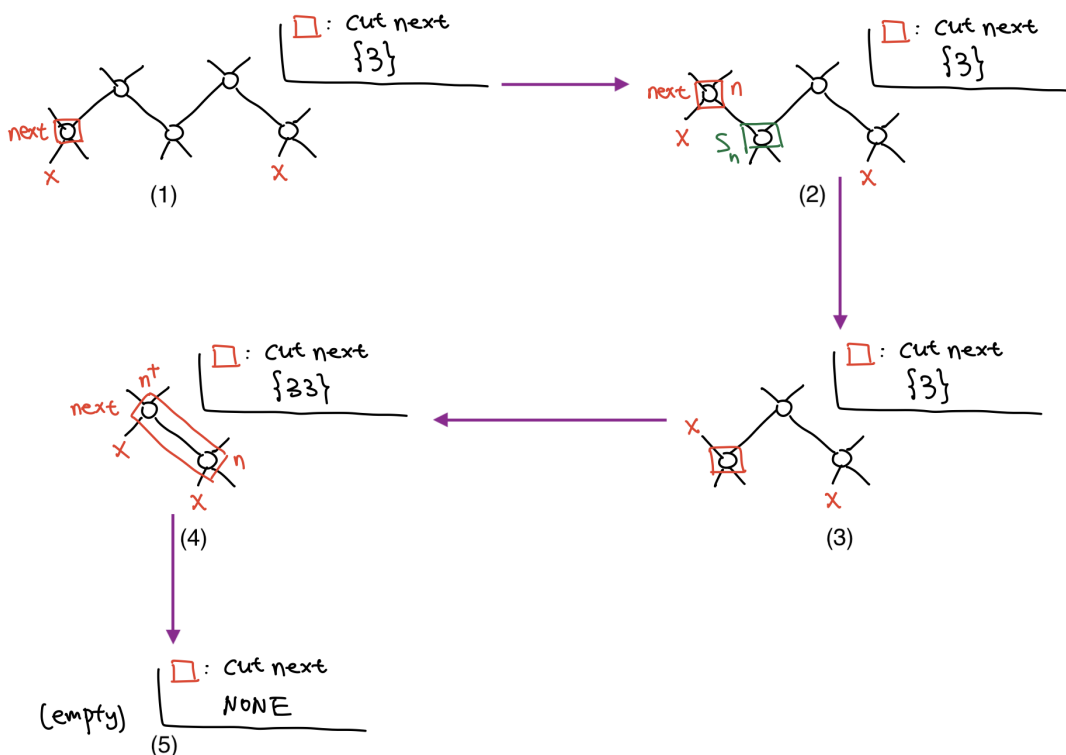


FIGURE 13. The **UP** algorithm. As in the above figures, the red box indicates the atom(s) to be cut next, the green box indicates the atom(s) in S_n (see Step (1) in the **UP** algorithm), which will be cut after cutting n . Note also when n and its parent n^+ (or child n^-) form a $\{33\}$ molecule, we should cut them together; see Step (2) of the **UP** algorithm.

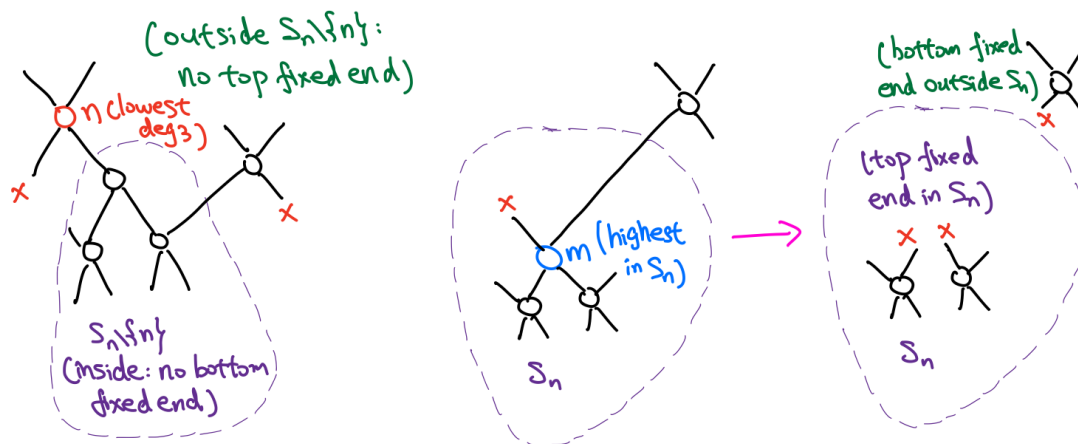


FIGURE 14. Illustration of the monotonicity property of the **UP** algorithm. Left: when we choose $\mathbf{n}$ in Step (1), there is no top fixed end outside $S_{\mathbf{n}} \setminus \{\mathbf{n}\}$ and no fixed end at all in $S_{\mathbf{n}} \setminus \{\mathbf{n}\}$ (i.e. all deg 4). Right: when we cut a highest remaining atom $\mathbf{m}$ in $S_{\mathbf{n}}$, this only creates top fixed end on $S_{\mathbf{n}}$ and bottom fixed end outside $S_{\mathbf{n}}$.

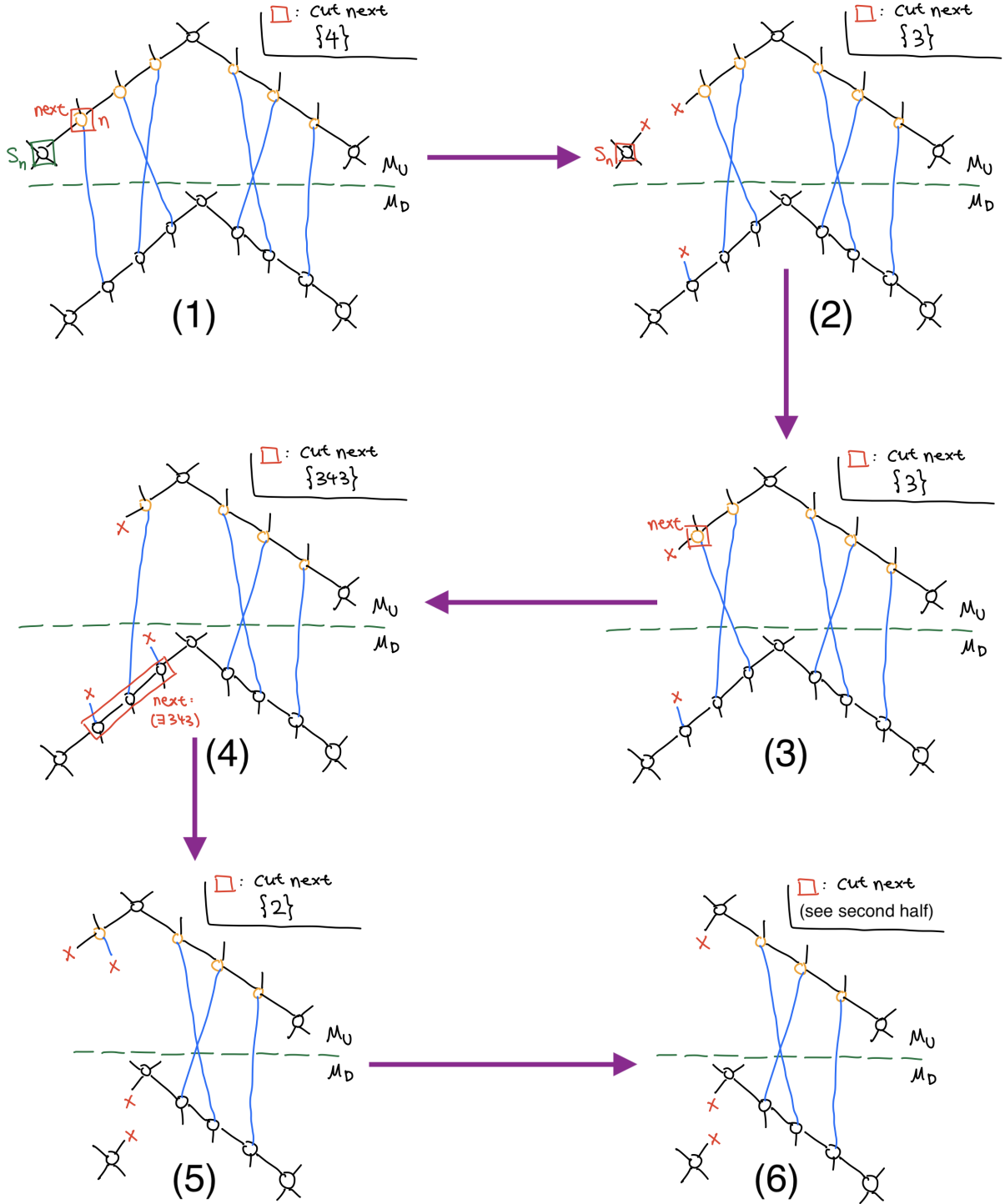


FIGURE 15. Algorithm for toy model I plus (first half) in Definition 4.3, applied to the example in Figure 10. In addition to Figure 11, we use green box to indicate atom(s) in S_n (see Step (2) in the toy model I plus algorithm) which will be cut after cutting n .

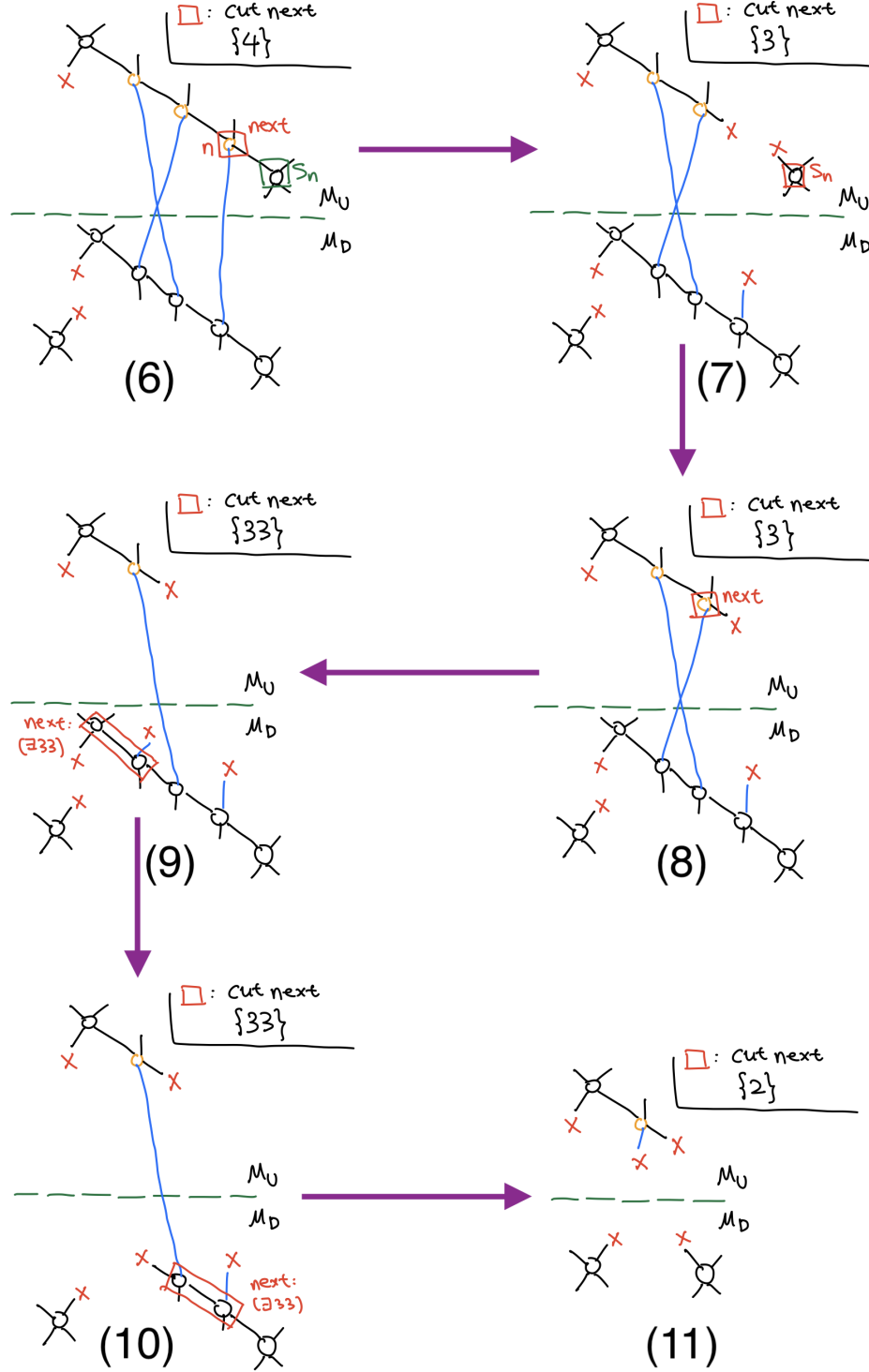


FIGURE 16. Algorithm for toy model I plus (second half). Here, the next atom n we choose is not the unique deg 3 atom in M_U , but rather the atom in X below it, see Step (1) in the toy model I plus algorithm. This leads to another $\{4\}$ loss, but the total loss is bounded by $\#_{\text{comp}}(X) = 2$. In the end, we subsequently cut two deg 2 atoms in M_U , and two deg 3 atoms in M_D , to conclude.

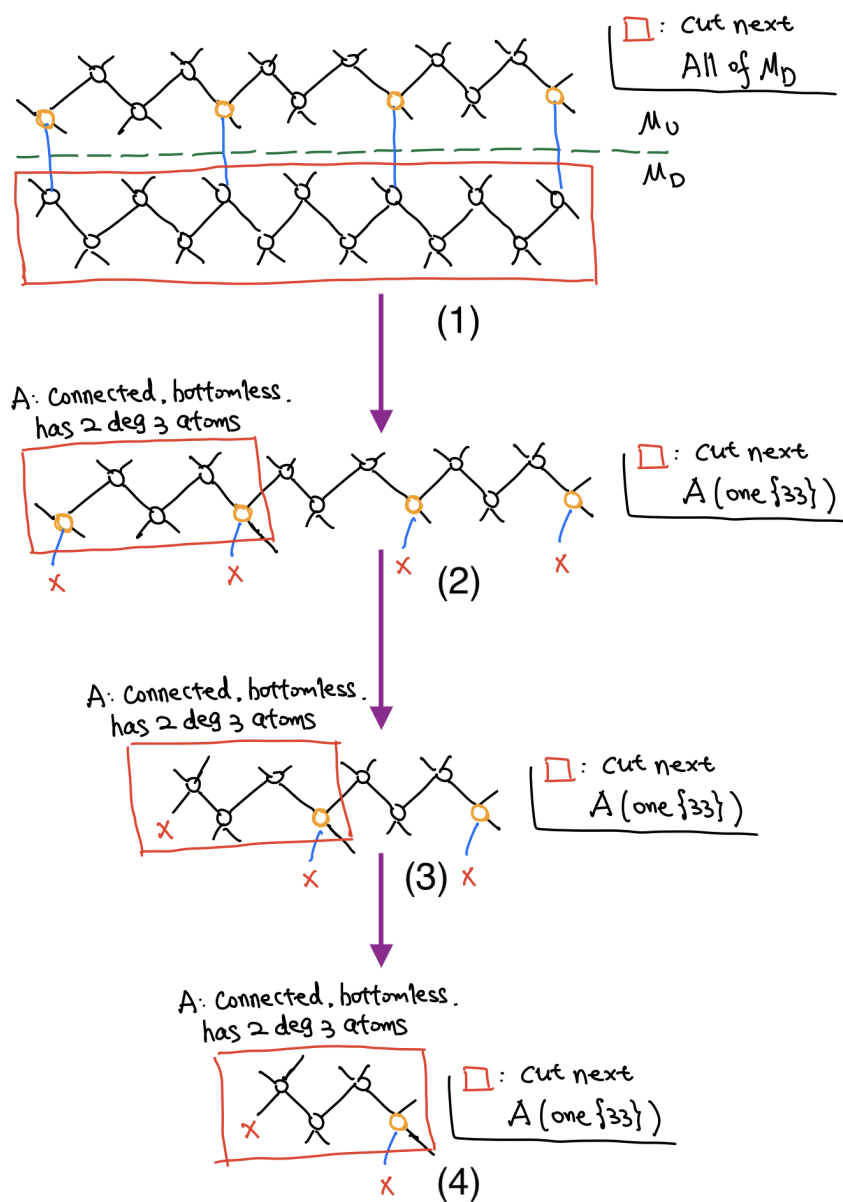


FIGURE 17. Algorithm for toy model II, applied to the example in Figure 10. Here we first cut all of M_D as free. Then, each time we choose a connected, no-bottom subset A that contains exactly two deg 3 atoms, see Step (1) in the toy model II algorithm. We then apply **UP** to cut all atoms in A (cf. Figure 13 for the application of **UP**) and repeat. Note that each application of **UP** guarantees at least one $\{33\}$ molecule.

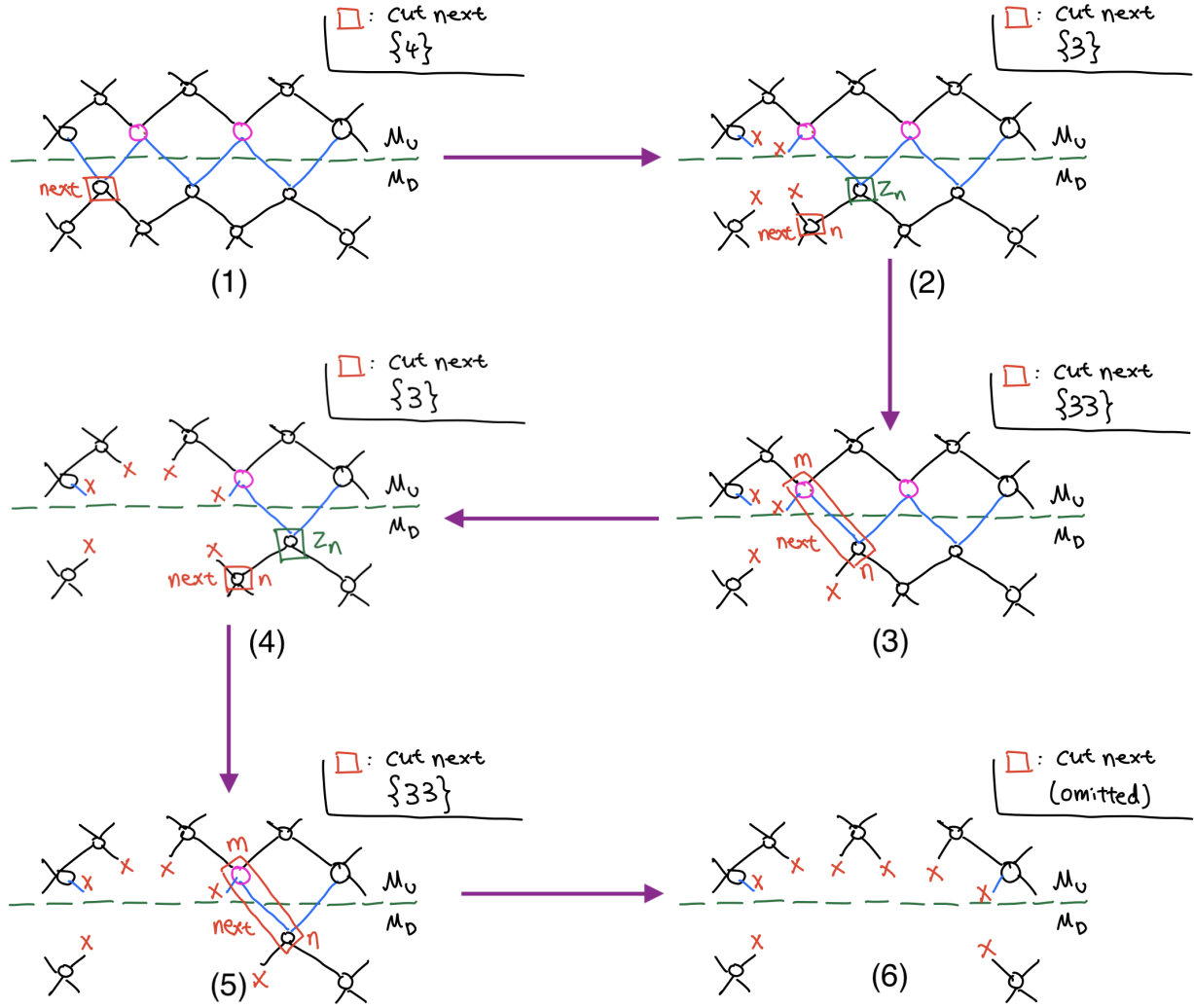


FIGURE 18. Algorithm for toy model III, applied to the example in Figure 10. Here we perform **DOWN** which is the dual of **UP** in $\mathbb{M}_D$, so in Step (1) in **UP** algorithm we choose the *highest* deg 3 atom instead of the lowest, and in Step (2) in **UP** algorithm we choose the *lowest* remaining atom instead of the highest, etc. Here green box indicates atoms in Z_n (the set of *ancestors* of n), and when an atom $n \in \mathbb{M}_D$ forms a $\{33\}$ molecule with an atom $m \in \mathbb{M}_U$ we always cut it. In the end we see that each 2-connection (pink atom) belongs to one $\{33\}$ molecule, as desired. As the gain is already enough, the rest can be cut into elementary molecules using **UP** in $\mathbb{M}_U$ and **DOWN** in $\mathbb{M}_D$.

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